

Scientific Analysis of Moral Judgments using Tobit and Cluster-Aware Non-Parametric Robustness Models

Automated Research Pipeline

marzo 30, 2026

1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from the Bucaramanga Campus. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator’s decision resulted in varying degrees of payoff for themselves, their own group, and a victim group. Under Option 2: judgment-level relational modeling, each judgment is treated as relational and linked both to a descriptive victim x negotiator scenario shorthand and to hypothesis-specific predictors for negotiator-side structure, observer-side victim alignment when applicable, and additional relational controls.

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{isjr}	Observed moral judgment for participant i , scenario s , judged negotiator j , and role r .
y_{isjr}^*	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
$\beta_{0.5}$	Median-regression coefficient from the non-parametric censored robustness model.
$Q_{0.5}(y_{isjr}^* \mathbf{x}_{isjr})$	Conditional median of the latent bounded outcome given the predictors.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
$CaseConfig_{isjr}$	Judgment-level relational shorthand retained only for backward compatibility with older descriptive artifacts.
$\mathbf{S}_{isj}$	Negotiator-side structure block encoding the judged and counterpart negotiators jointly within a judgment row.
V_{is}	Observer-side player-victim outgroup indicator used only in the bystander H2 specification.
G_{isjr}	Judged negotiator relational status (ingroup, outgroup, or control) defined relative to the role-relevant reference actor for H3.
A_{is}	Decision indicator distinguishing Accept from Reject for H3.
C_{isjr}	Additional controls, including participant demographics, slot effects, and the remaining relational terms not absorbed by the main H2/H3 blocks.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

Option 2: judgment-level relational modeling retains a victim x judged-negotiator shorthand for descriptive summaries, while the executable hypotheses use explicit negotiator-level structure terms. H2 now uses the judged-plus-counterpart structure directly, and in observer rows it adds the player-victim alignment term and its interaction with that structure.

2.1 Interpretation of Interaction Terms

The models herein employ several predefined predictors. It is important to note how interaction terms are interpreted in the context of this behavioral experiment:

1. **Interaction Subsumption:** When an interaction term is statistically significant, it indicates that the effect of one variable depends on the level of the other. Crucially, if the interaction is significant but the constituent main effects are not explicitly significant, their effects are fully subsumed and contextualized by the interaction.
2. **Continuous by Discrete Interactions:** For terms like `iri_total:judged_outgroup`, a negative coefficient implies that the severity of moral judgment (lower score) induced by higher empathy is steeper (magnified) when evaluating an outgroup negotiator compared to an ingroup negotiator. A positive coefficient would mean empathy makes judgments less severe for the outgroup.
3. **Discrete by Discrete Interactions:** For terms like `judged_outgroup:decision_accept`, a positive coefficient implies that the change in moral judgment when moving from rejecting a deal to accepting a deal is more positive (less morally condemned) for an outgroup negotiator than for an ingroup negotiator.

3 Hypotheses to Test

- **H1 (Empathy under relational controls):** Within the victim and bystander subsets, higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on judged-negotiator status, counterpart status, decision outcome, observer-side victim alignment when applicable, and participant controls.
- **H2 (Negotiator-side relational structure):** Moral-judgment severity will differ as a function of the judgment-level ingroup/outgroup/control structure of the judged and counterpart negotiators, with an additional player-victim alignment interaction in the bystander subset.
- **H3 (Empathy x judged-status moderation):** The empathy effect may vary across judged-negotiator relational status, while decision outcome and relational controls remain in the model.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, negotiator-specific moral judgments (y_{isjr}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y_{isjr}^*) that follows a linear relationship. Option 2 replaces isolated ingroup/outgroup indicators with explicit judgment-level relational variables built from the paired-group structure of each judgment. The analytic hypothesis models decompose that structure into judged-negotiator status, counterpart-negotiator status, observer-side victim alignment when applicable, and hypothesis-specific interaction blocks. H1 retains accepted-sample victim x judged-negotiator case contrasts. H2 now uses subset-specific relational-structure blocks, and H3 retains the judged-status, decision, and empathy-interaction design:

Victim-subset H2:

$$y_{isj,Victim}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(V)} + \delta' \mathbf{Z}_i + \epsilon_{isj}$$

Bystander-subset H2:

$$y_{isj,Obs}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(O)} + \eta V_{is} + \boldsymbol{\theta}'(\mathbf{S}_{isj}^{(O)} \times V_{is}) + \boldsymbol{\delta}' \mathbf{Z}_i + \epsilon_{isj}$$

H3:

$$y_{isjr}^* = \beta_0 + \beta_1 \text{Empathy}_i + \beta_2' G_{isjr} + \beta_3 A_{is} + \beta_4' (G_{isjr} \times A_{is}) + \beta_5' (\text{Empathy}_i \times G_{isjr}) + \beta_6' C_{isjr} + \epsilon_{isjr}, \quad \epsilon_{isjr} \sim N(0, \sigma^2)$$

The actual observed judgment y_{isjr} relates to this latent variable via the censoring transformation:

$$y_{isjr} = \max(-9, \min(9, y_{isjr}^*))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

Because the Tobit model relies on a Gaussian latent-error assumption, the pipeline also fits a distribution-robust censored median specification implemented as interval-censored quantile regression ($p = 0.5$). This complementary estimator targets the conditional median of the latent bounded outcome and is less sensitive to heavy tails and non-normal disturbances:

$$Q_{0.5}(y_{isjr}^* \mid \mathbf{x}_{isjr}) = \mathbf{x}_{isjr}' \beta_{0.5}$$

The non-parametric branch preserves the censoring structure while relaxing the parametric normality assumption. The Tobit and robustness results should therefore be interpreted jointly: Tobit provides the clustered parametric benchmark with participant-clustered standard errors by id, while the non-parametric branch first establishes a converged full-sample censored median fit and then, in the default pipeline, adds participant-level cluster bootstrap inference that resamples ids with replacement and retains all repeated observations from each sampled participant. In both branches, id is used only to account for within-participant dependence and is not a substantive explanatory variable.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors for Tobit and participant-level cluster bootstrap inference for the non-parametric robustness model after each converged full-sample fit, we ensure that both sets of p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.096, with an Effective Sample Size (ESS) of 1405.4.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

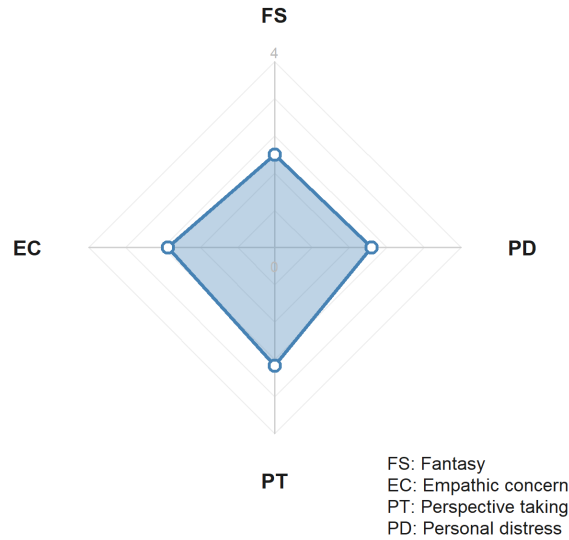
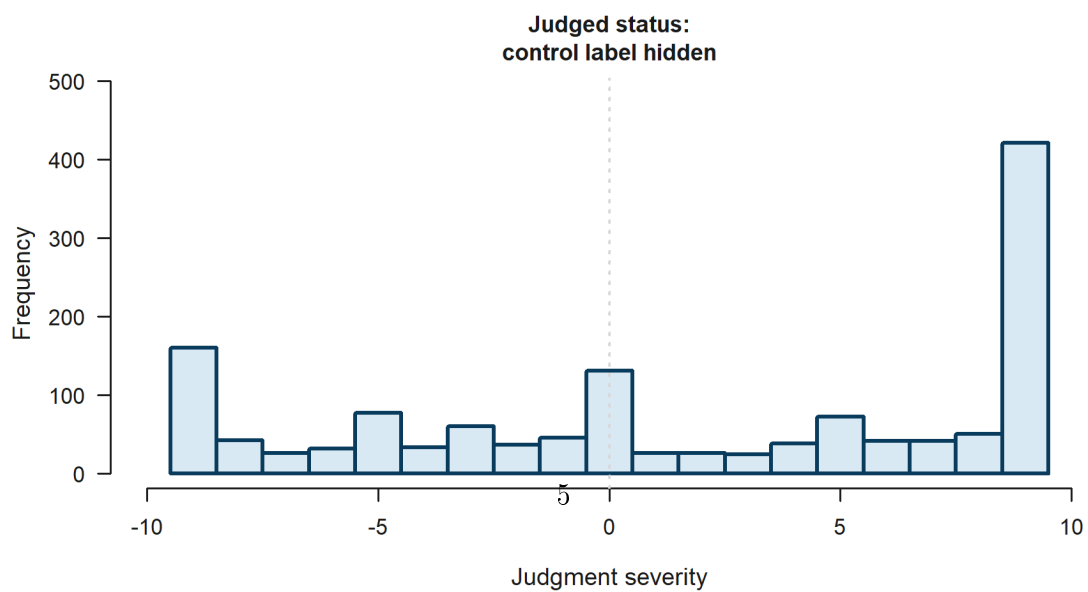
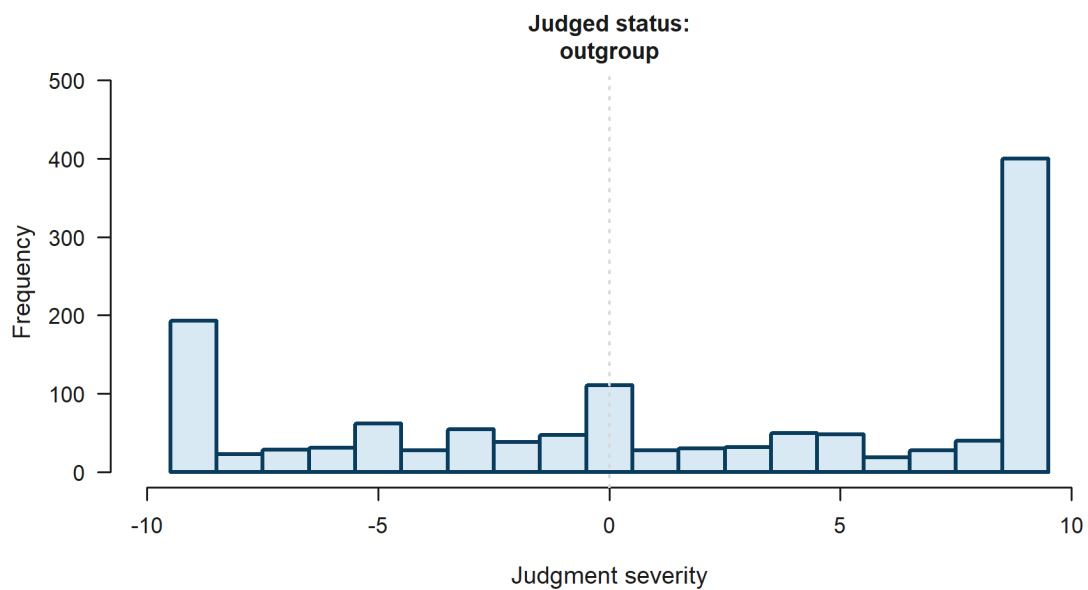
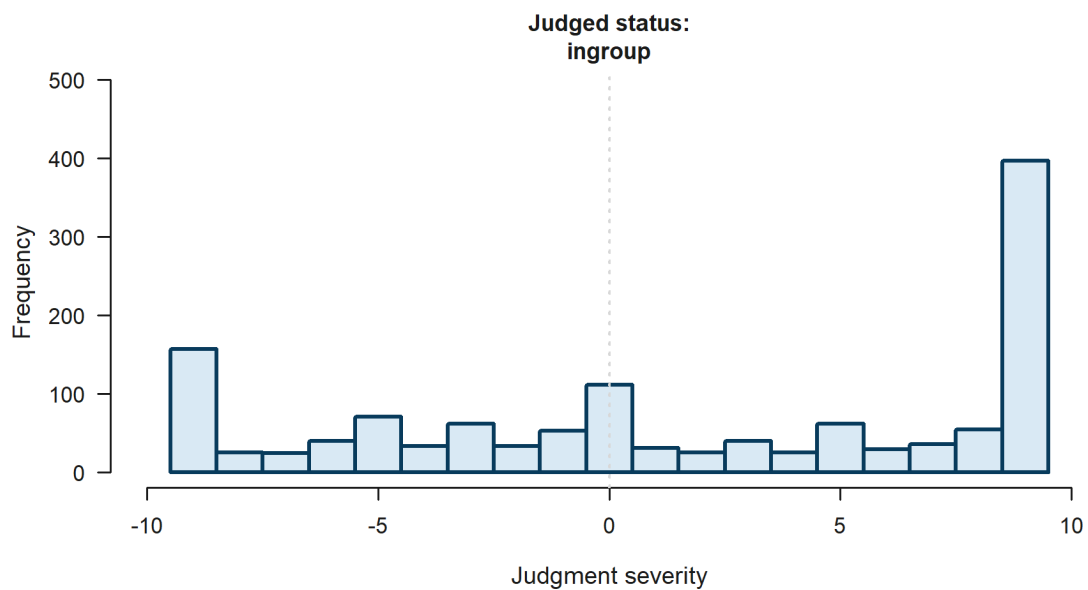


Figure 1: IRI Latent Variable Averages (Radar Plot profile).

Figure 2 summarizes the overall judgment distribution across judged-negotiator status, and Figures 3 and 4 replicate that severity-panel logic across the six explicit victim x negotiator case configurations within the victim and bystander subsets.



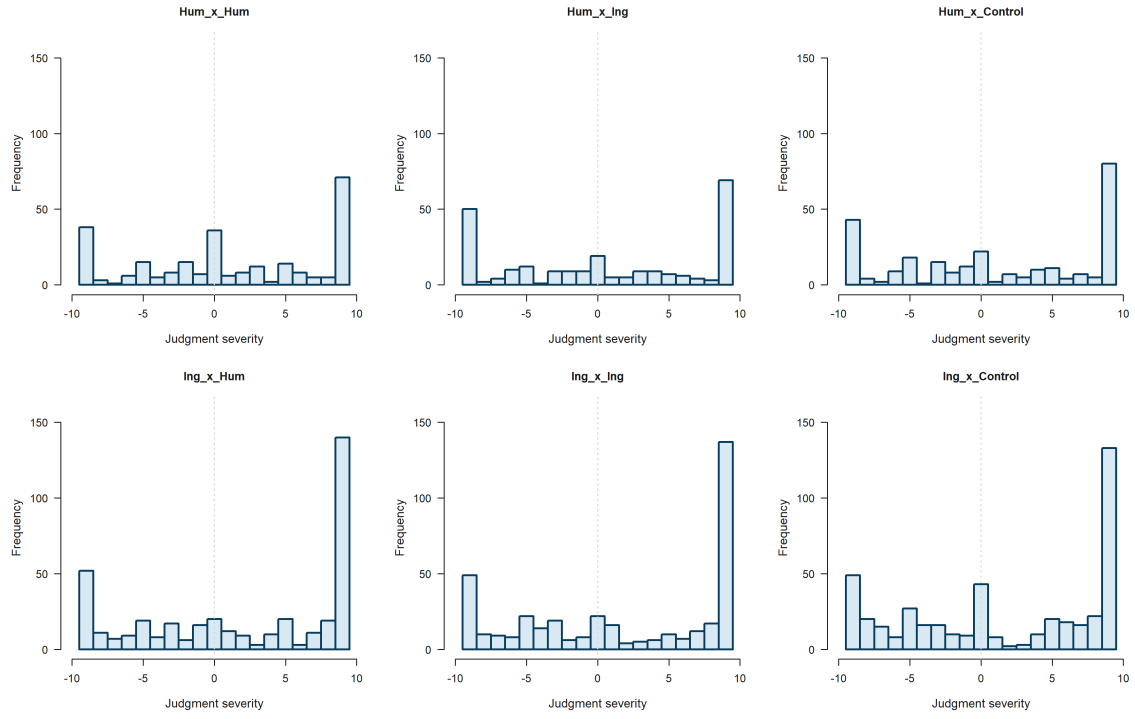


Figure 3: Victim-subset severity panels across the six explicit victim x negotiator case configurations, using a fixed judgment scale from -10 to 10.

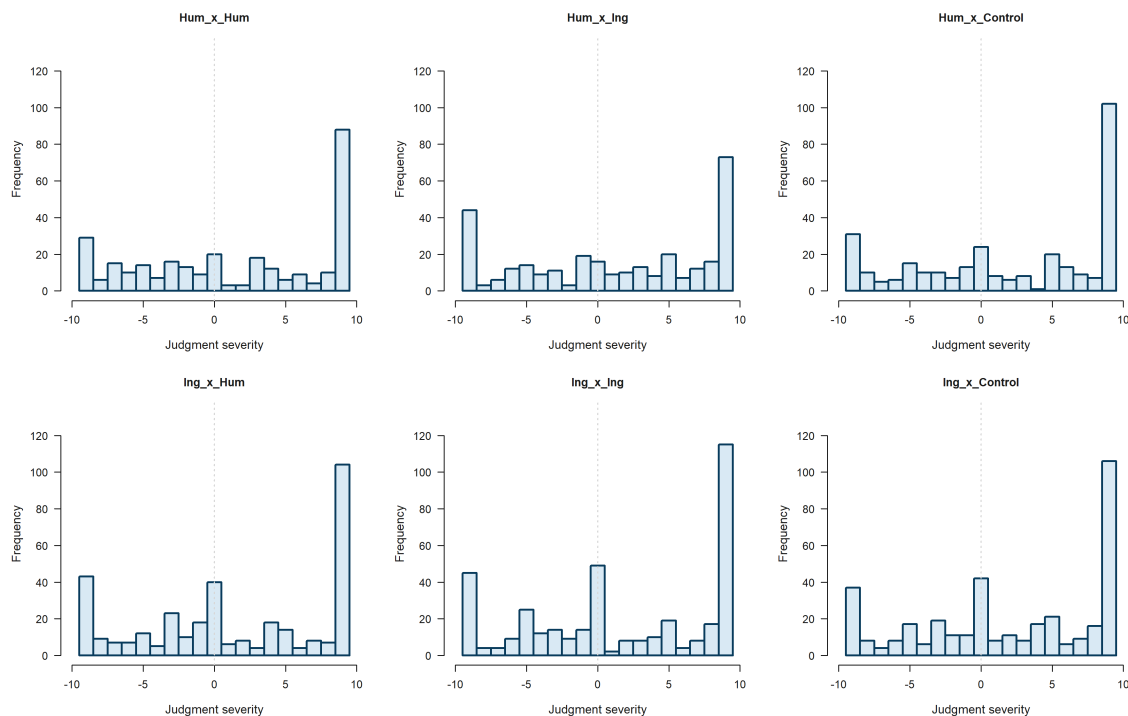


Figure 4: Bystander-subset severity panels across the six explicit victim x negotiator case configurations, using a fixed judgment scale from -10 to 10.

The descriptive branch no longer centers the main report on victim x negotiator case-label tables; the hypothesis tests below work directly with negotiator-level relational predictors.

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.500	0.554	0.509	0.830	-0.057
0.500	1.000	0.424	0.649	0.818	-0.173
0.554	0.424	1.000	0.301	0.730	0.098
0.509	0.649	0.301	1.000	0.762	-0.009
0.830	0.818	0.730	0.762	1.000	-0.049
-0.057	-0.173	0.098	-0.009	-0.049	1.000

Visual representations of these relationships, including fitted lines with 95% confidence bands, are provided in Figure 5.

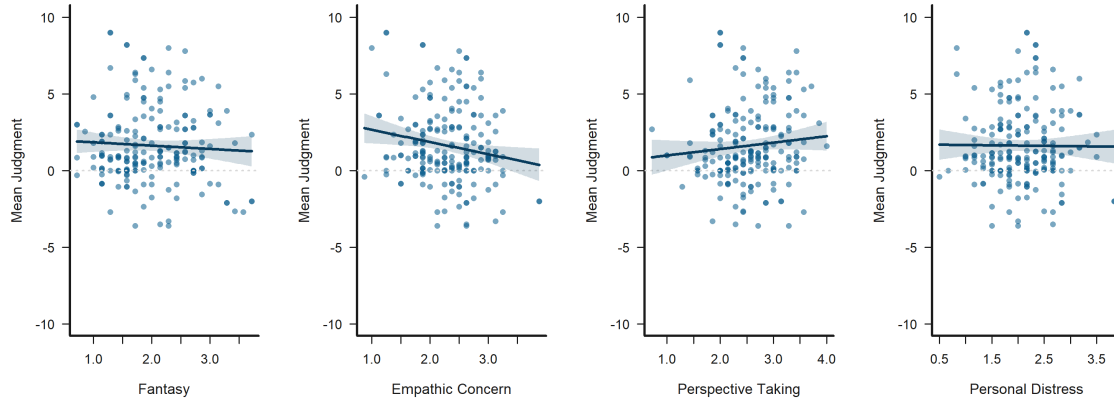


Figure 5: Bivariate Scatters: IRI Scales vs. Mean Judgment, with fitted lines and 95% confidence bands.

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models used in this report.

Table 3: Estimator fit summary across Tobit specifications.

Model	Approach	Status	Iterations	Observations	Participants	ClusterUnit
H1_A_Bystander_Total_Tobit	Tobit	completed	5	1990	199	id
H1_A_Victim_Total_Tobit	Tobit	completed	6	1990	199	id
H1_B_Bystander_Constructs_Tobit	Tobit	completed	6	1990	199	id
H1_B_Victim_Constructs_Tobit	Tobit	completed	6	1990	199	id
H2_A_Bystander_Total_Tobit	Tobit	completed	3	1990	199	id
H2_A_Victim_Total_Tobit	Tobit	completed	3	1990	199	id
H2_B_Bystander_Constructs_Tobit	Tobit	completed	3	1990	199	id
H2_B_Victim_Constructs_Tobit	Tobit	completed	3	1990	199	id
H3_A_Bystander_Total_Tobit	Tobit	completed	6	1990	199	id
H3_A_Victim_Total_Tobit	Tobit	completed	6	1990	199	id
H3_B_Bystander_Constructs_Tobit	Tobit	completed	6	1990	199	id
H3_B_Victim_Constructs_Tobit	Tobit	completed	6	1990	199	id

8.1 Hypothesis Significance Summary

The following concise tables list only hypothesis-relevant predictors that reached at least $p < 0.10$ in the available Tobit models, split by victim and bystander subsets, using conventional symbols to indicate strength of evidence (+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Dynamic figures are generated only for predictors that appear in these tables with at least one significance symbol.

Table 4: Hypothesis-level significance summary for the victim subset across Tobit models.

Hypothesis	Tobit support
H1	Empathy: Perspective taking**; Empathy: Empathic concern**; Empathy: Personal distress+
H2	None
H3	Personal distress x judged negotiator outgroup*; Empathic concern x judged negotiator outgroup*; Empathy x judged negotiator outgroup*; Empathy x judged negotiator control label hidden+

Victim subset

Table 5: Hypothesis-level significance summary for the bystander subset across Tobit models.

Hypothesis	Tobit support
H1	Empathy: Perspective taking*; Empathy: Fantasy scale*
H2	Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)*; Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)+
H3	Empathy x judged negotiator control label hidden**; Personal distress x judged negotiator outgroup*

Bystander subset

8.2 Significance-Driven Figures

Only hypothesis-relevant predictors that reach $p < 0.10$ or better are visualized automatically. These dynamic figures use the saved Tobit outputs, and participant id remains only an inference-level clustering unit rather than a substantive explanatory variable.

H1: Empathy under relational controls Empathy: Perspective taking is statistically significant in the Tobit model (**, $p = 0.003$). Figure 6 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

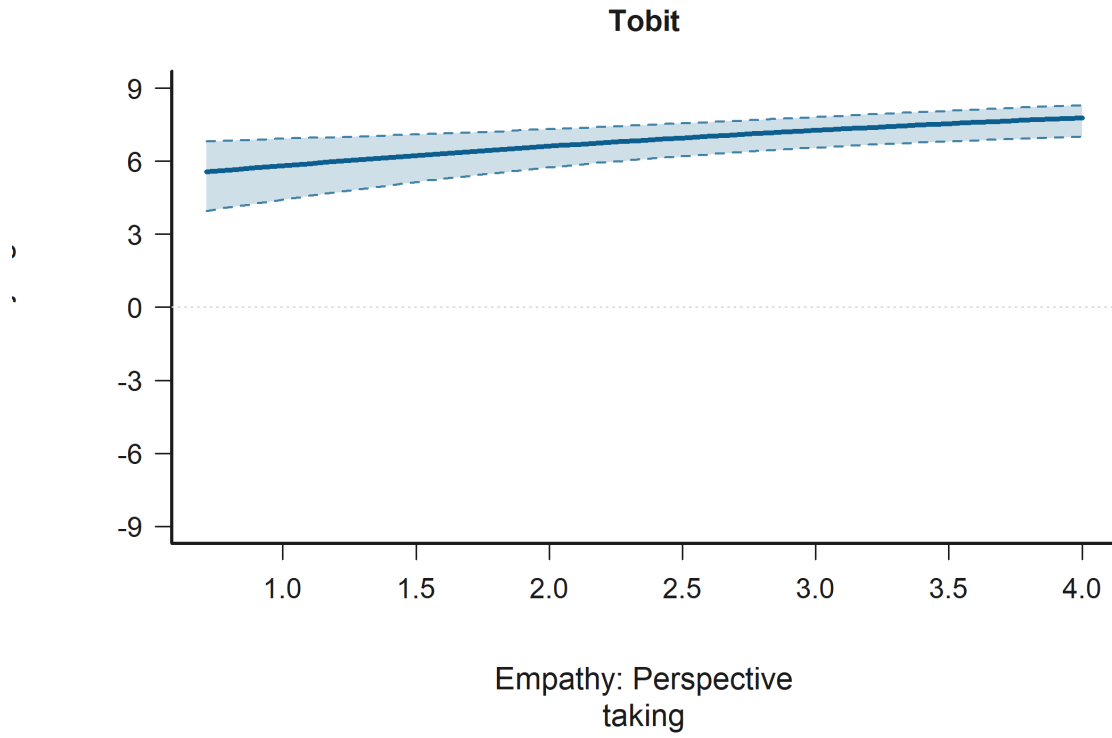


Figure 6: Effect Plot for Empathy: Perspective taking in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in the Tobit model (**, $p = 0.004$). Figure 7 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted judgment.

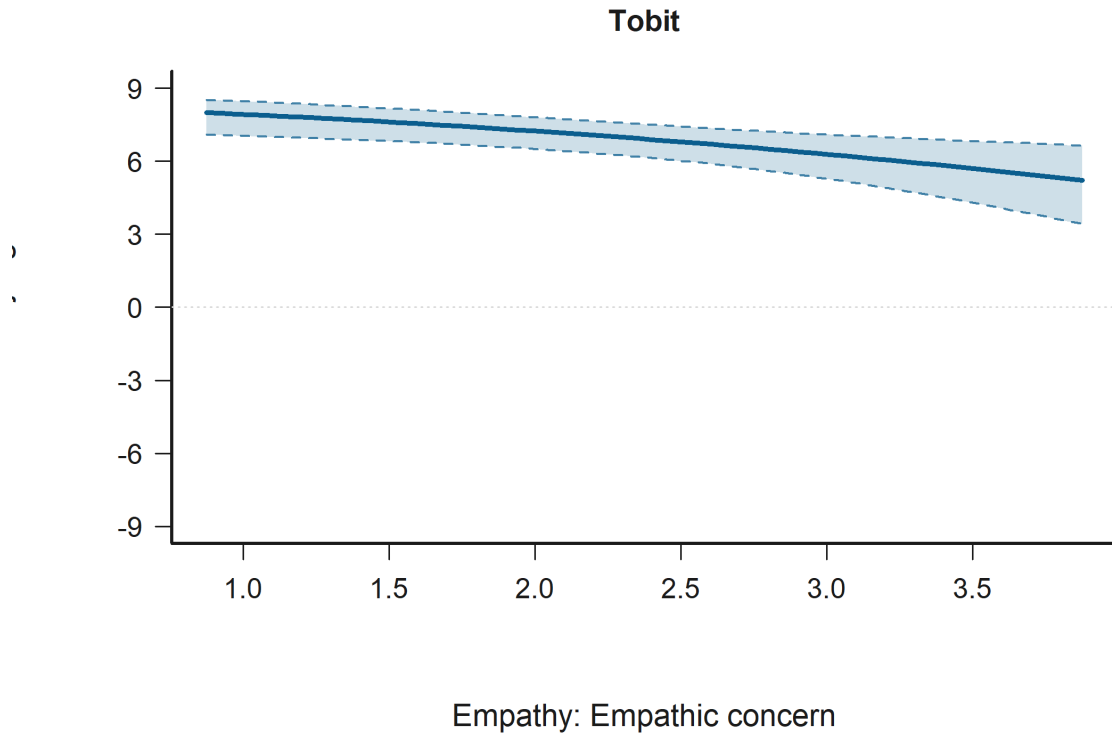


Figure 7: Effect Plot for Empathy: Empathic concern in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Fantasy scale is statistically significant in the Tobit model (*, $p = 0.042$). Figure 8 shows that across the observed range, higher Empathy: Fantasy scale corresponds to lower predicted judgment.

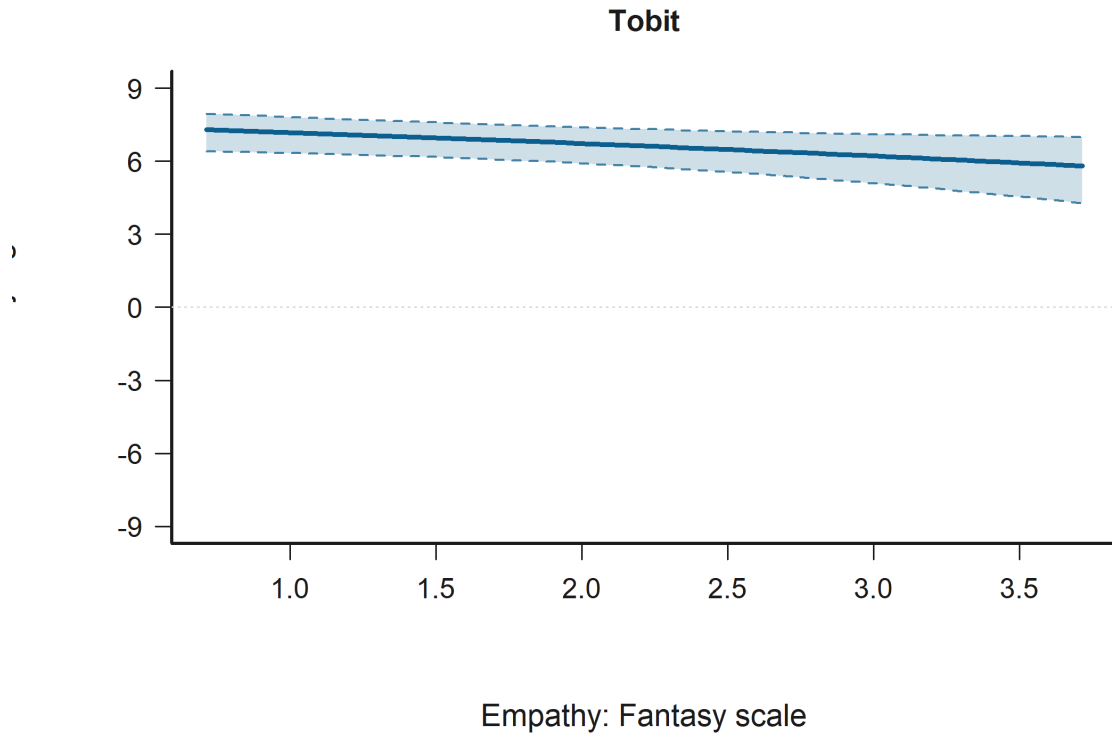


Figure 8: Effect Plot for Empathy: Fantasy scale in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Personal distress is statistically significant in the Tobit model (+, $p = 0.053$). Figure 9 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted judgment.

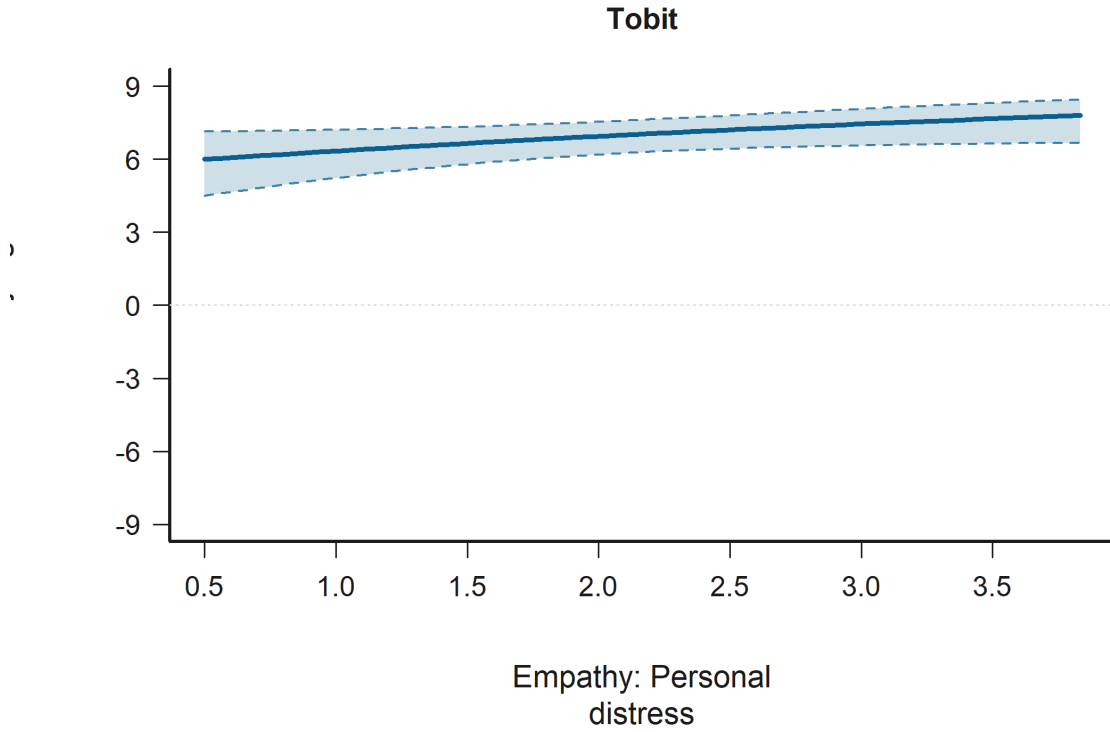


Figure 9: Effect Plot for Empathy: Personal distress in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

H2: Negotiator-side relational structure Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim) is statistically significant in the Tobit model (*, $p = 0.030$). Figure 10 shows that the predicted relationship rises most sharply for Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) when the condition is Player and victim outgroup-aligned.

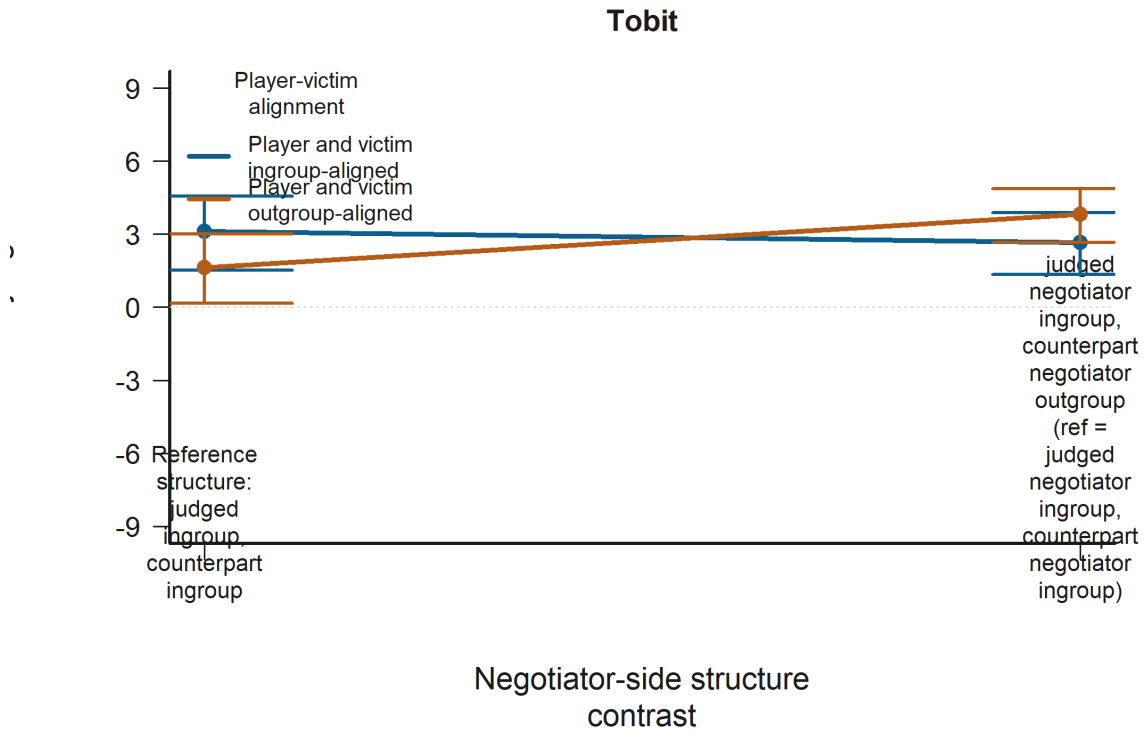


Figure 10: Interaction Plot for Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim) in H2: Negotiator-side relational structure. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) is statistically significant in the Tobit model (+, $p = 0.057$). Figure 11 shows that predicted judgment is lower for judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) than for Reference structure: judged ingroup, counterpart ingroup.

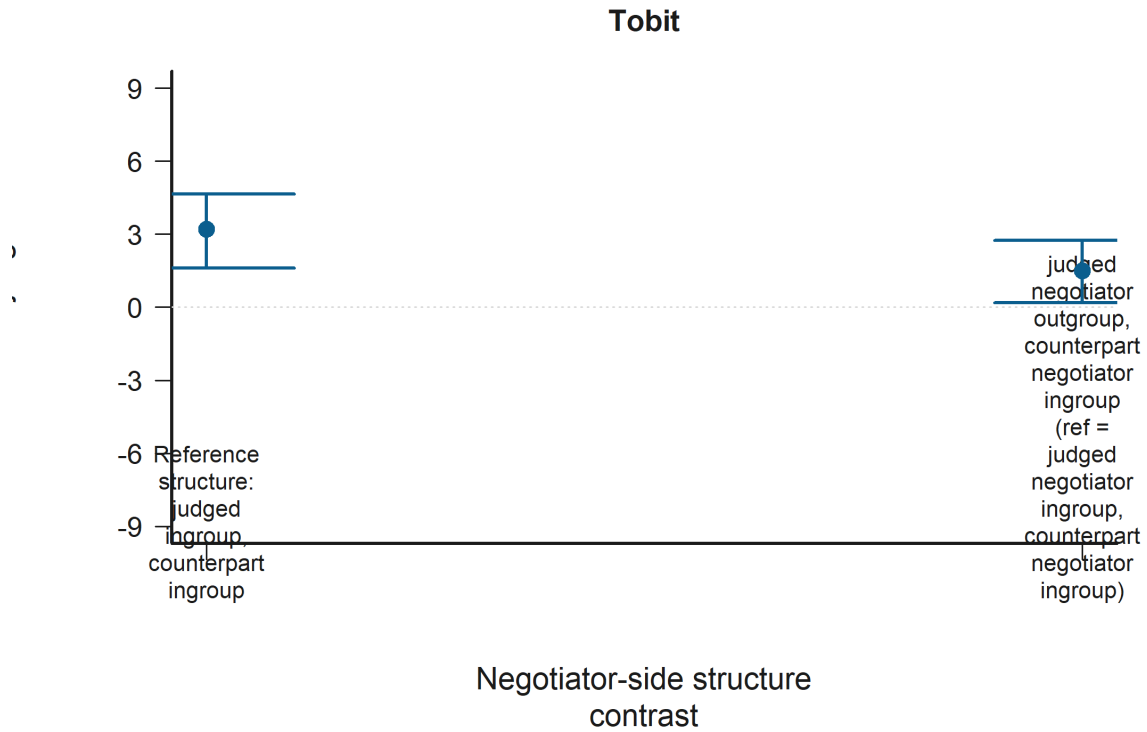


Figure 11: Grouped Prediction Plot for Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) in H2: Negotiator-side relational structure. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

H3: Empathy x judged-status moderation Empathy x judged negotiator control label hidden is statistically significant in the Tobit model (**, $p = 0.007$). Figure 12 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is Judged negotiator control label hidden.

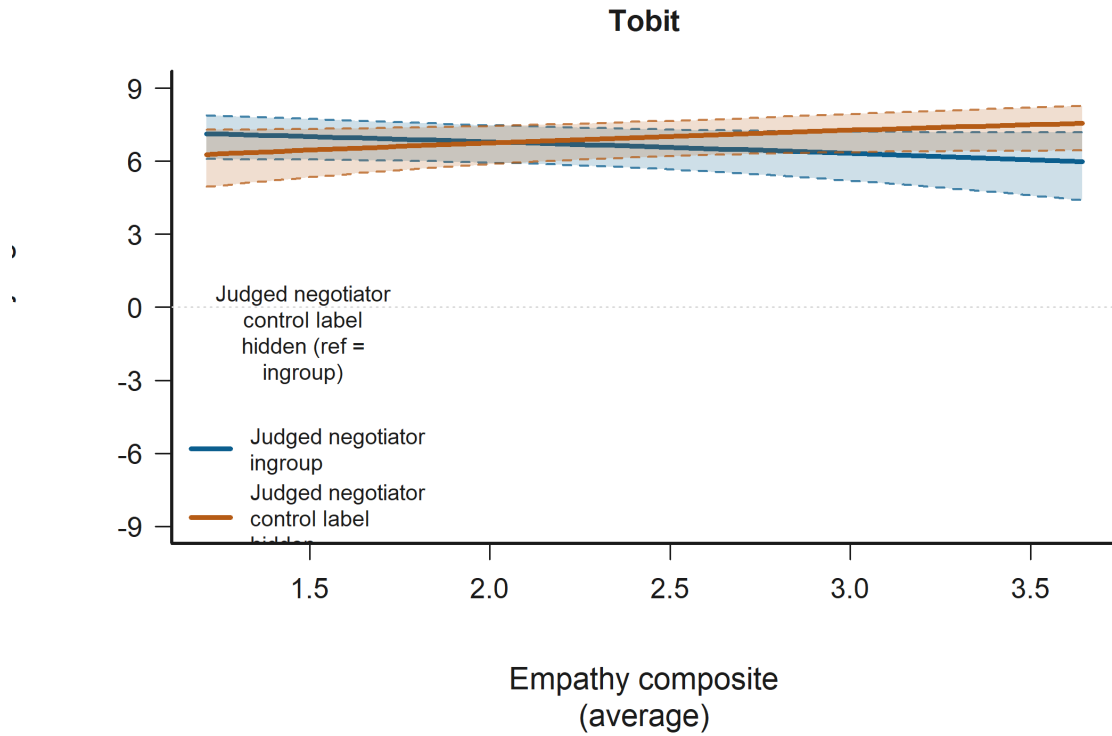


Figure 12: Interaction Plot for Empathy x judged negotiator control label hidden in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Personal distress x judged negotiator outgroup is statistically significant in the Tobit model (*, $p = 0.016$). Figure 13 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Judged negotiator outgroup.



Figure 13: Interaction Plot for Personal distress x judged negotiator outgroup in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathic concern x judged negotiator outgroup is statistically significant in the Tobit model (*, $p = 0.026$). Figure 14 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Judged negotiator outgroup.

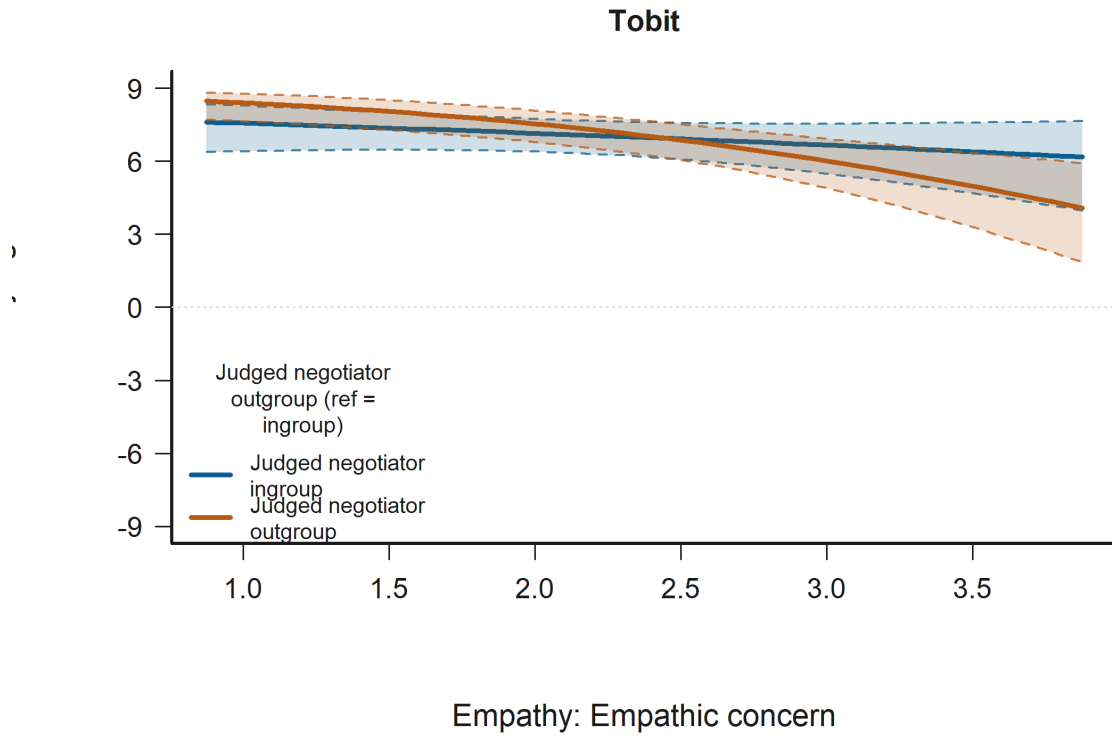


Figure 14: Interaction Plot for Empathic concern x judged negotiator outgroup in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy x judged negotiator outgroup is statistically significant in the Tobit model (*, $p = 0.050$). Figure 15 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is Judged negotiator ingroup.

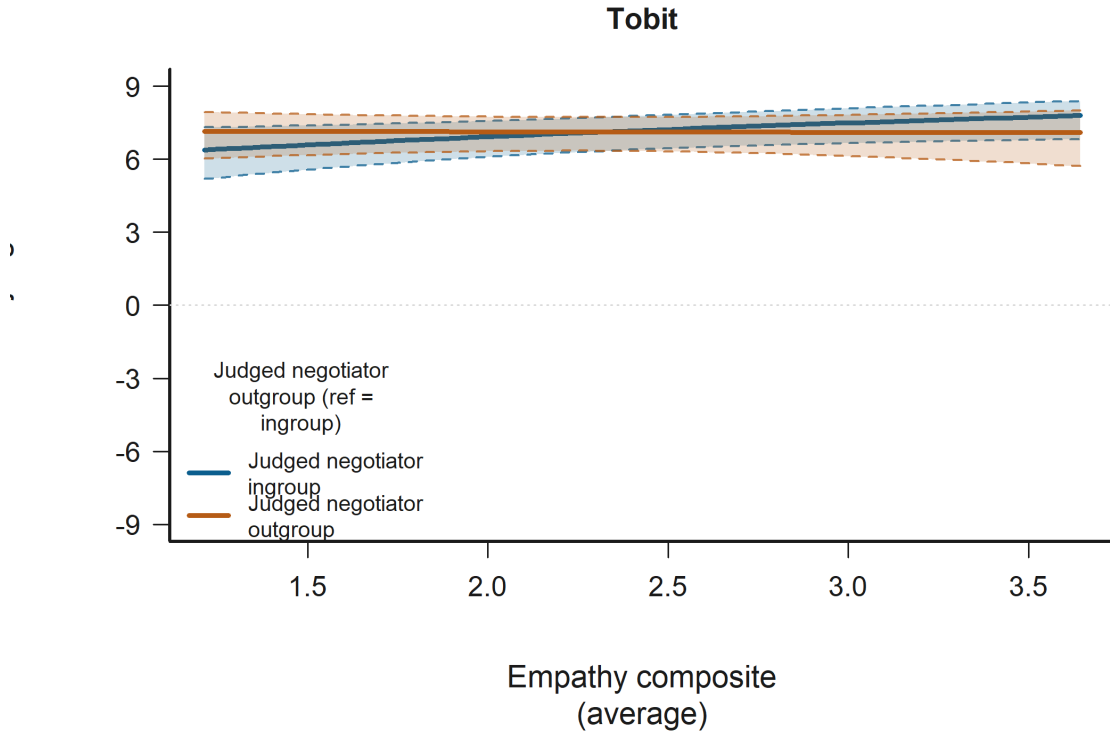


Figure 15: Interaction Plot for Empathy x judged negotiator outgroup in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

8.3 All Significant Predictors ($p < 0.10$)

The following figures extend beyond the hypothesis-target terms and visualize every predictor that reaches $p < 0.10$ in the available H1-H3 Tobit models. This includes significant controls such as age when they clear the threshold.

judgments_bystander Negotiator accepted harmful deal is statistically significant in:

- H1: Empathy under relational controls Model A (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 16 shows that predicted judgment is lower for Accept harmful deal than for Reject harmful deal.



Figure 16: Grouped Prediction Plot for Negotiator accepted harmful deal in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy x judged negotiator control label hidden is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 17 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is Judged negotiator control label hidden.

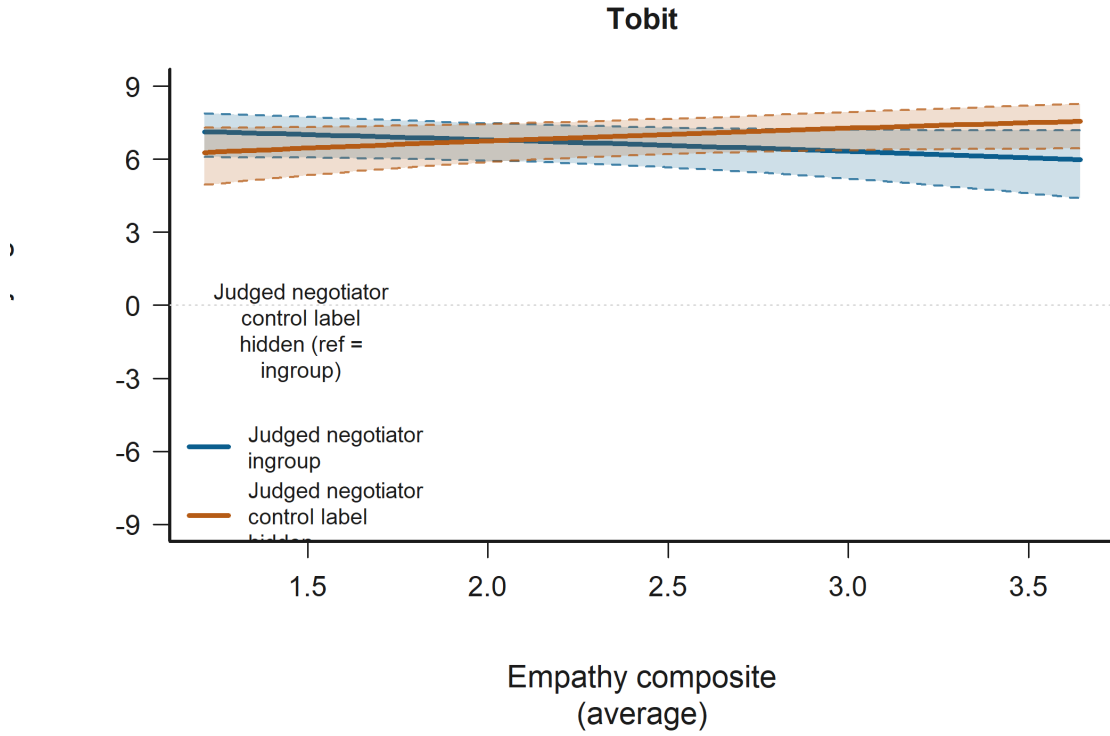


Figure 17: Interaction Plot for Empathy x judged negotiator control label hidden in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Socioeconomic status is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 18 shows that predicted judgment is higher for Socioeconomic status 5 than for Socioeconomic status 0.

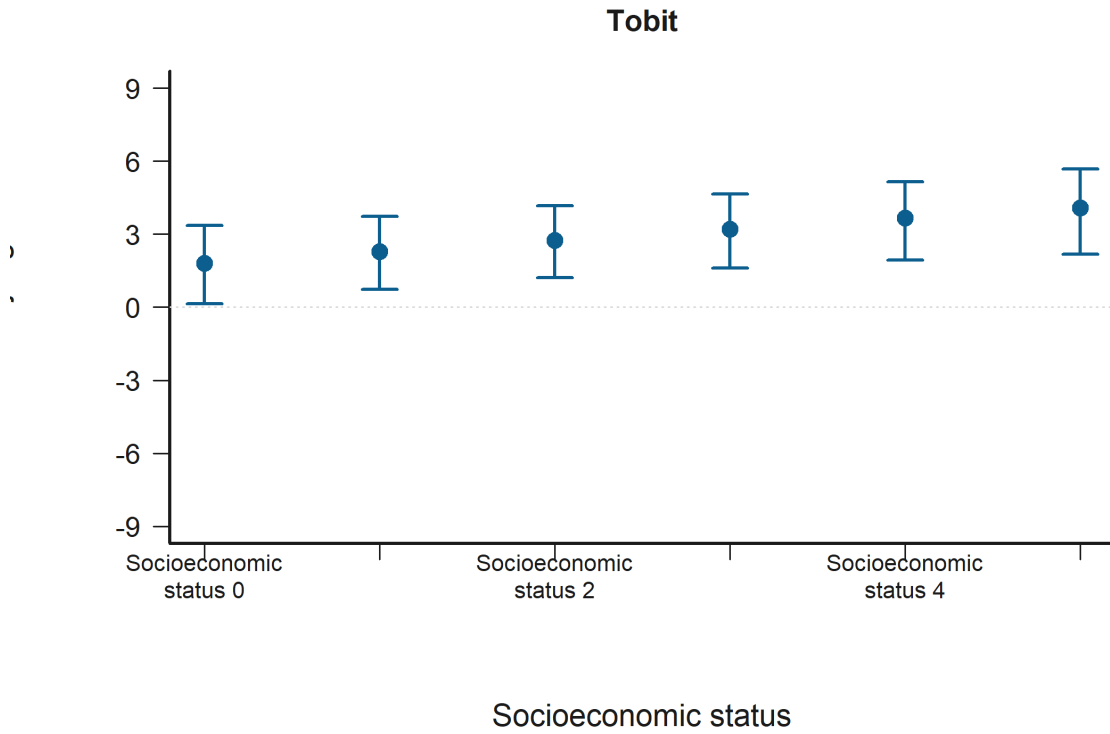


Figure 18: Grouped Prediction Plot for Socioeconomic status in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Judged negotiator control label hidden (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 19 shows that predicted judgment is higher for Judged negotiator control label hidden than for Judged negotiator ingroup.



Figure 19: Grouped Prediction Plot for Jugged negotiator control label hidden (ref = ingroup) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Personal distress x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 20 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Jugged negotiator outgroup.



Figure 20: Interaction Plot for Personal distress x judged negotiator outgroup in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Engineering participant (ref = humanities) is statistically significant in:

- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H2: Negotiator-side relational structure Model A (Tobit)

Figure 21 shows that predicted judgment is higher for Engineering participant than for Humanities participant.

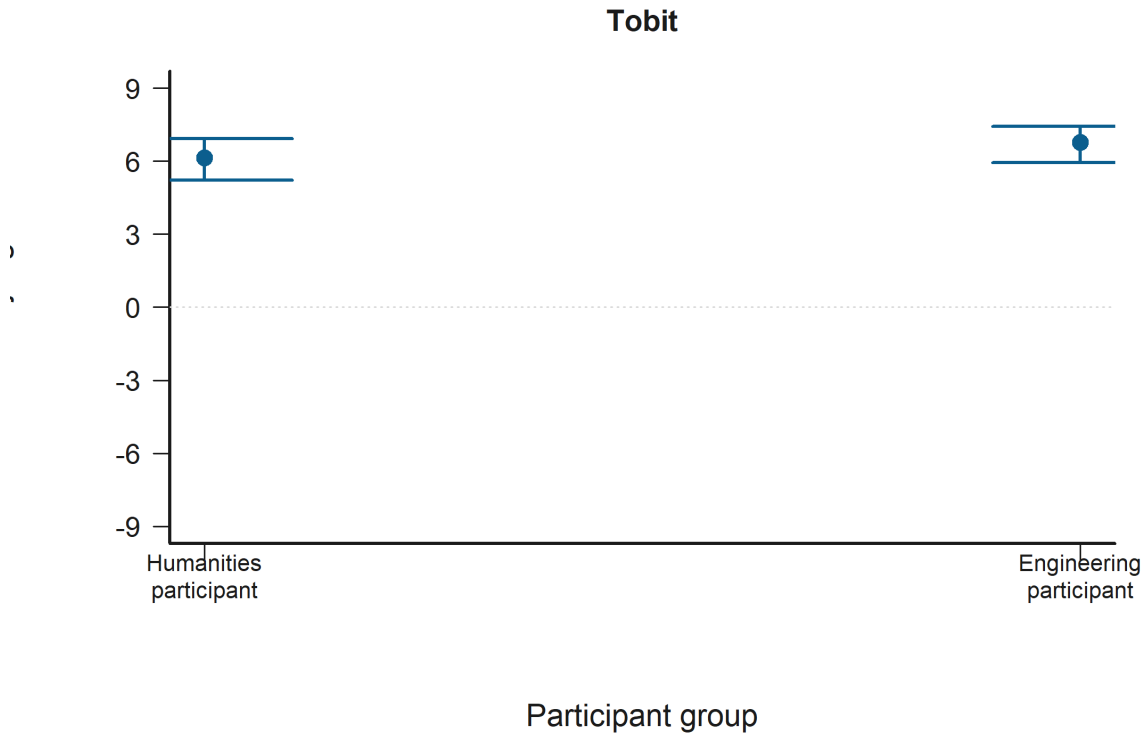


Figure 21: Grouped Prediction Plot for Engineering participant (ref = humanities) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Counterpart negotiator control label hidden (ref = ingroup) is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 22 shows that predicted judgment is higher for Counterpart negotiator control label hidden than for Counterpart negotiator ingroup.

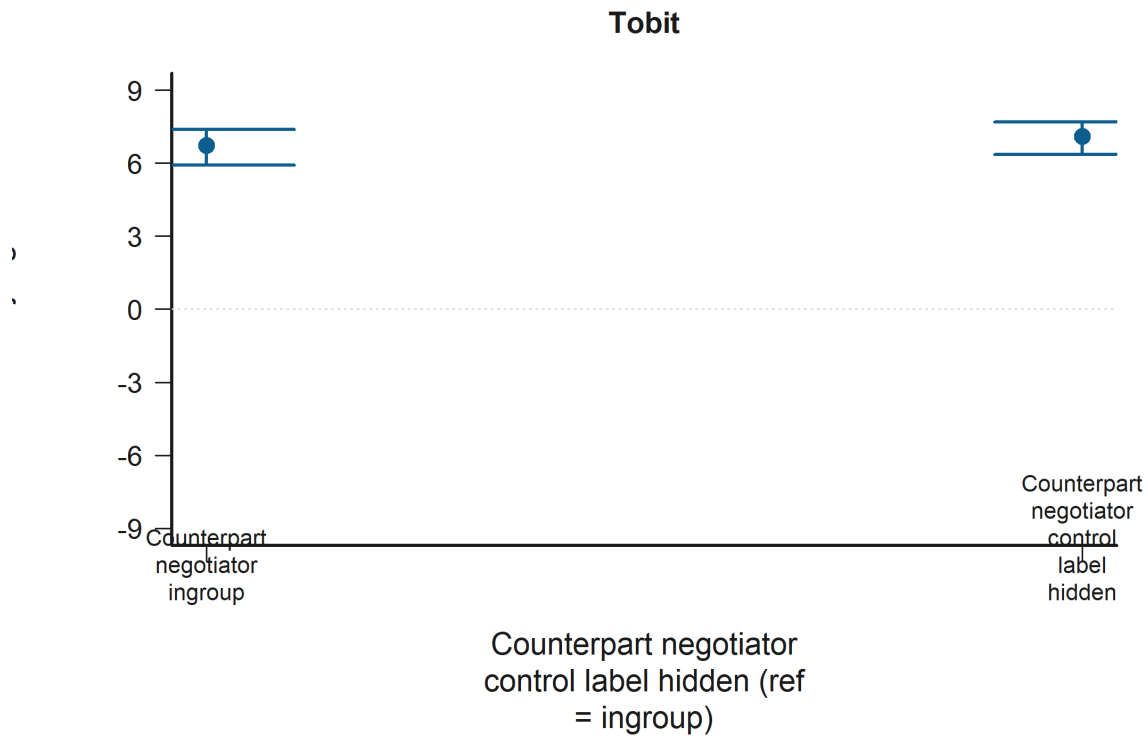


Figure 22: Grouped Prediction Plot for Counterpart negotiator control label hidden (ref = ingroup) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)

Figure 23 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted judgment.



Figure 23: Effect Plot for Empathy: Empathic concern in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Age is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 24 shows that across the observed range, higher Age corresponds to higher predicted judgment.

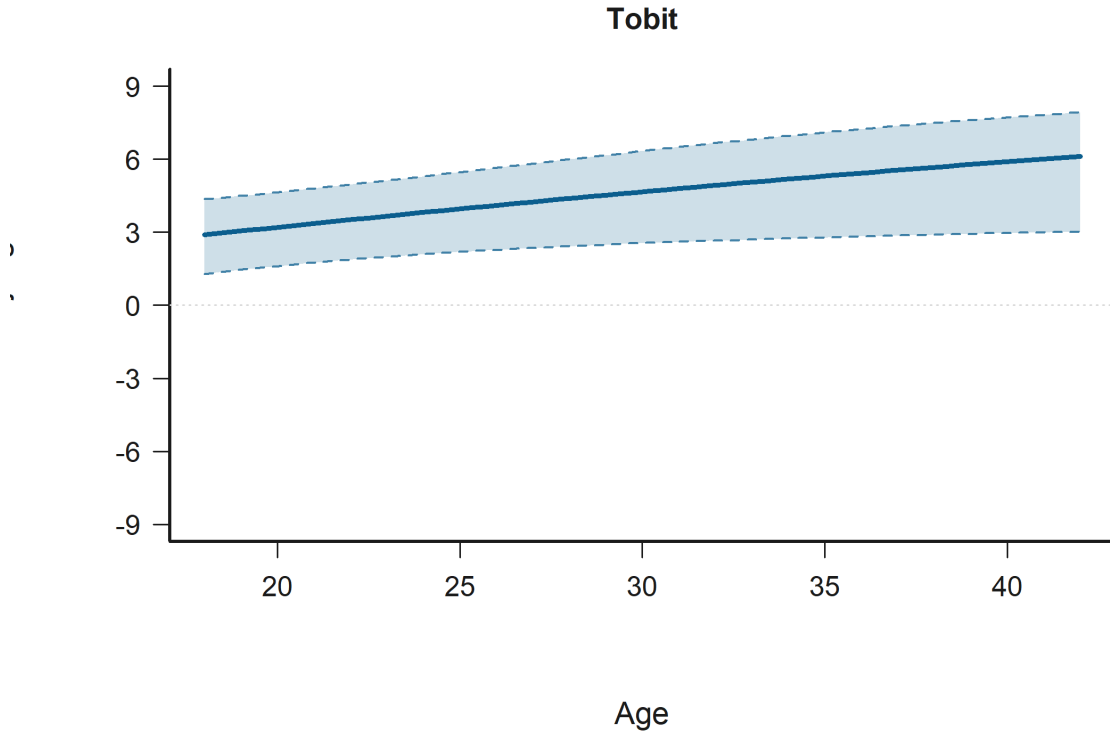


Figure 24: Effect Plot for Age in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim) is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H2: Negotiator-side relational structure Model A (Tobit)

Figure 25 shows that the predicted relationship rises most sharply for Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) when the condition is Player and victim outgroup-aligned.

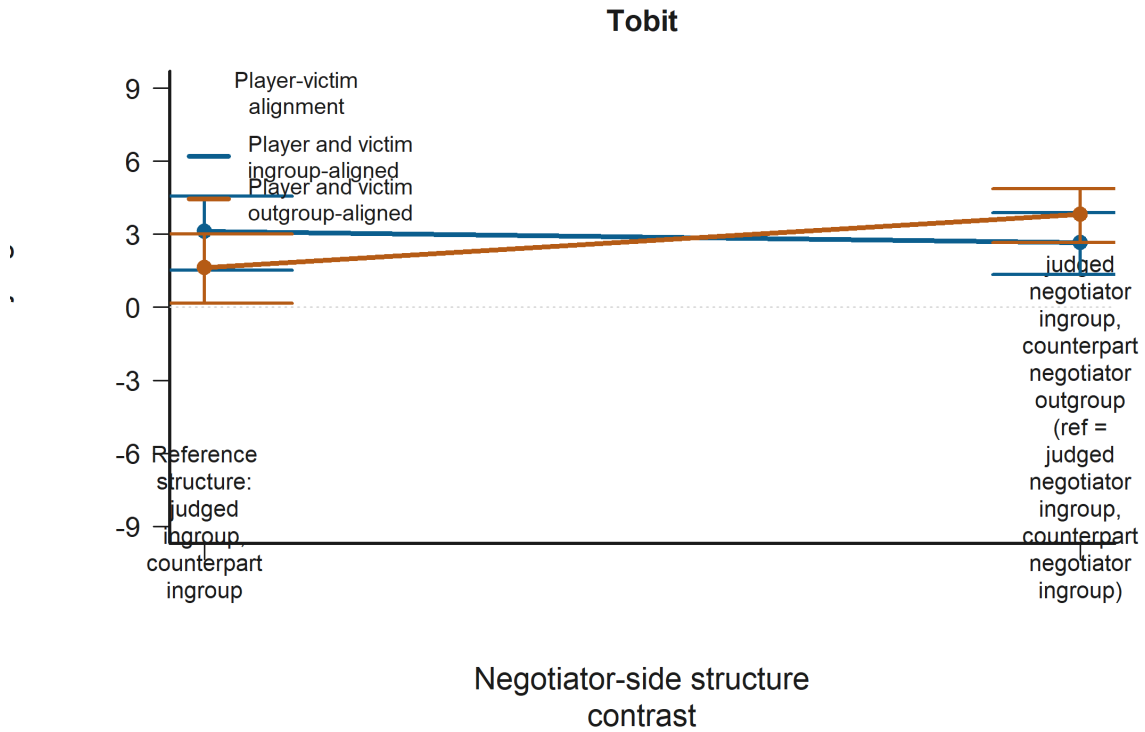


Figure 25: Interaction Plot for Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 26 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

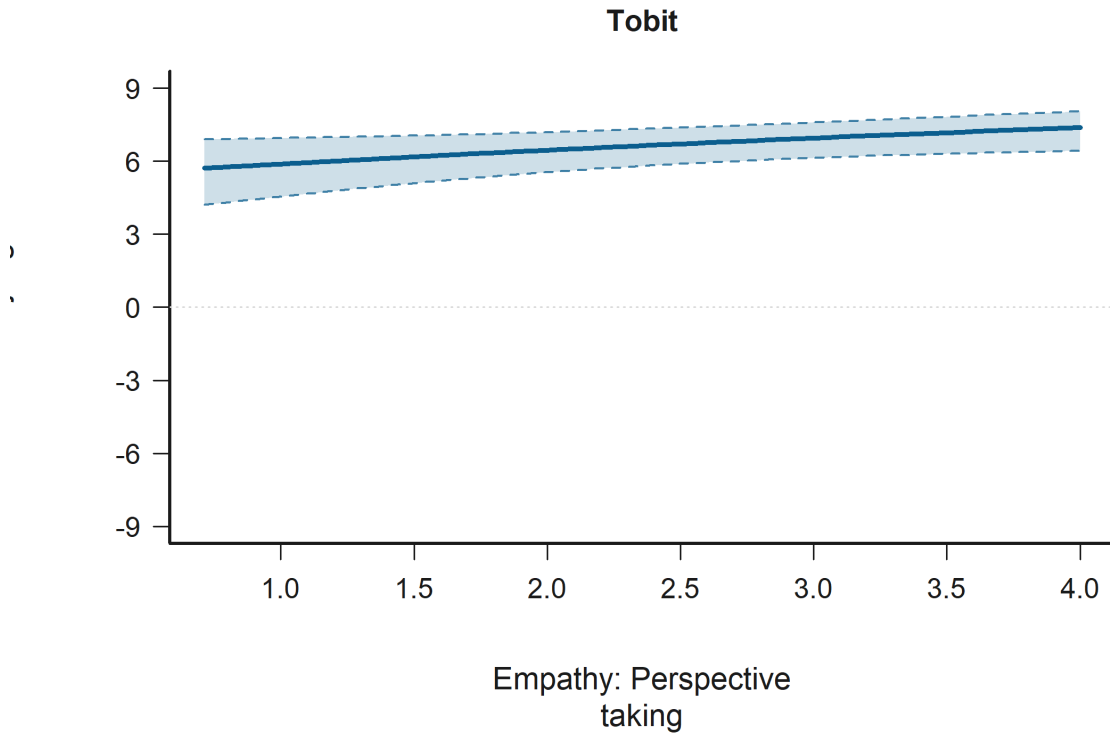


Figure 26: Effect Plot for Empathy: Perspective taking in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Fantasy scale is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 27 shows that across the observed range, higher Empathy: Fantasy scale corresponds to lower predicted judgment.

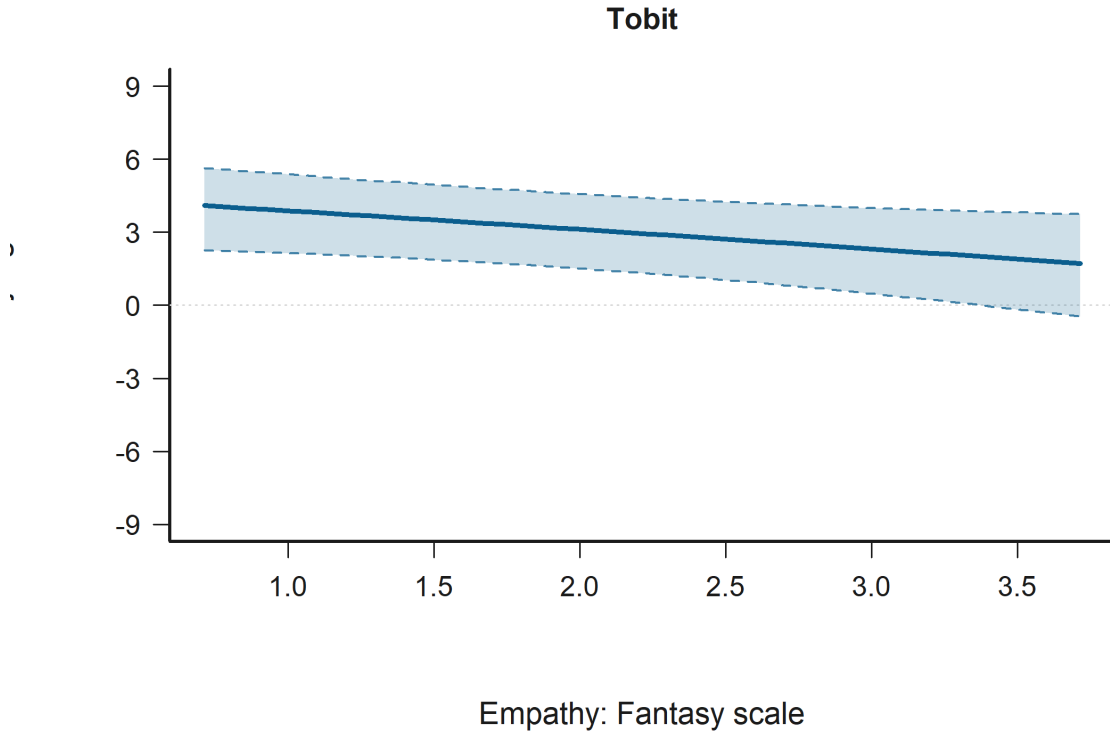


Figure 27: Effect Plot for Empathy: Fantasy scale in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 28 shows that predicted judgment is lower for judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) than for Reference structure: judged ingroup, counterpart ingroup.

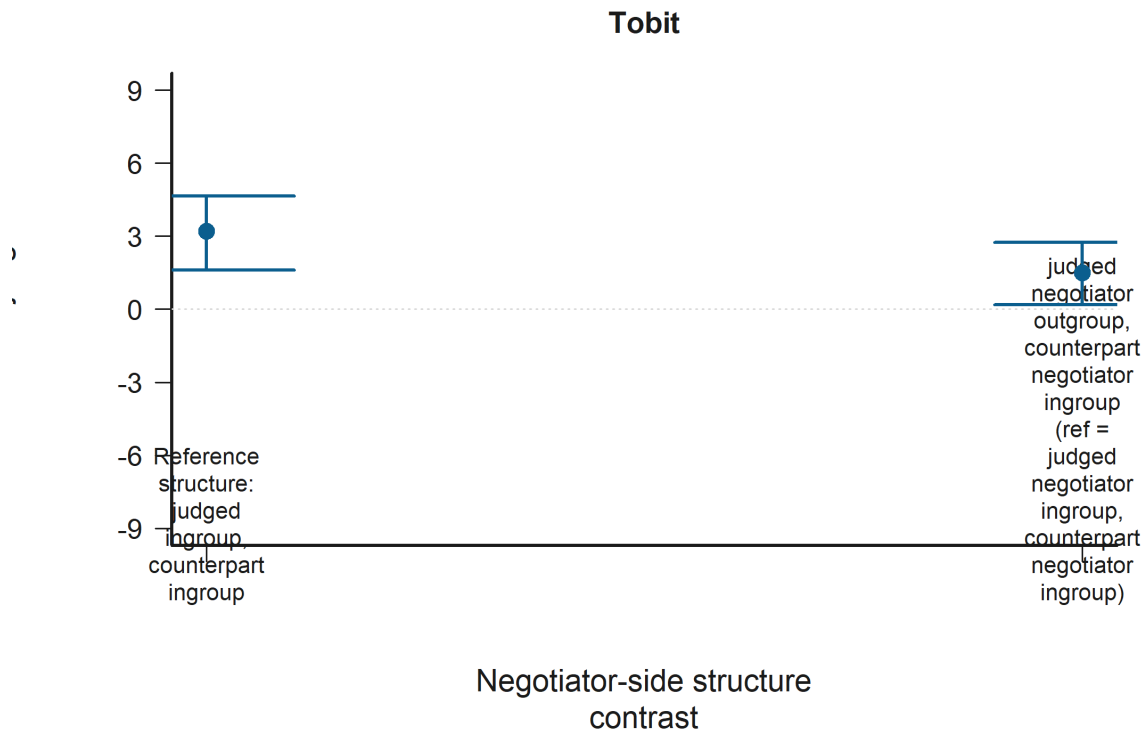


Figure 28: Grouped Prediction Plot for Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Personal distress is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)

Figure 29 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted judgment.

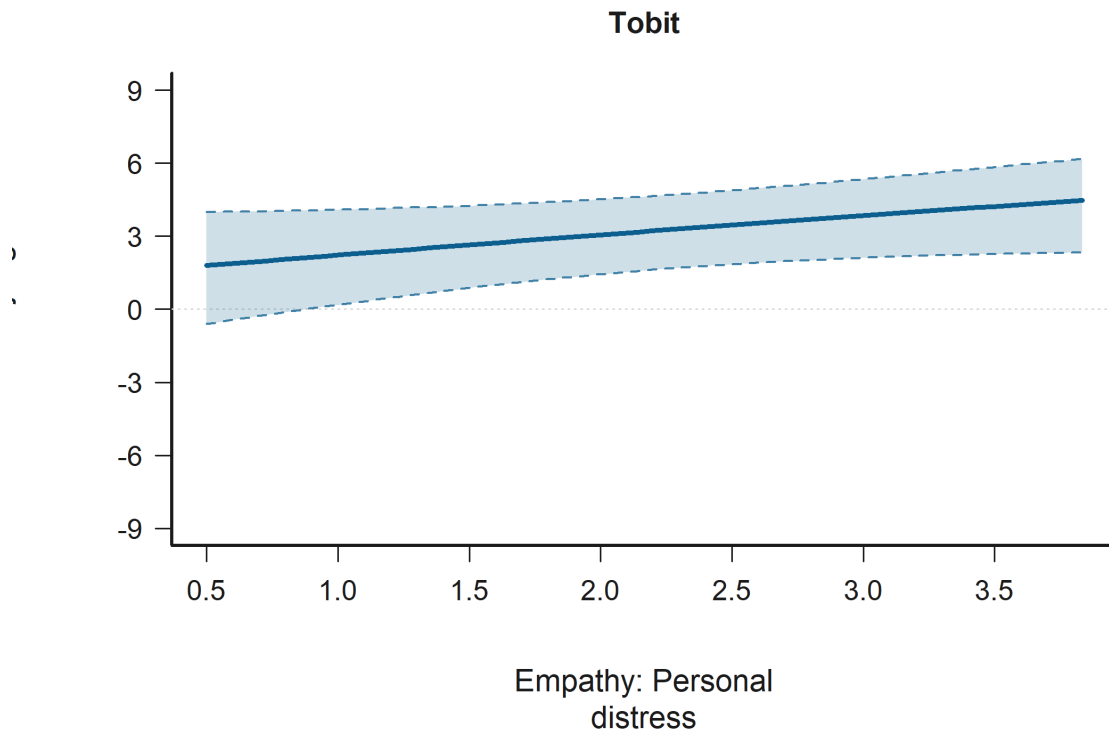


Figure 29: Effect Plot for Empathy: Personal distress in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy composite (average) is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)

Figure 30 shows that across the observed range, higher Empathy composite (average) corresponds to lower predicted judgment.

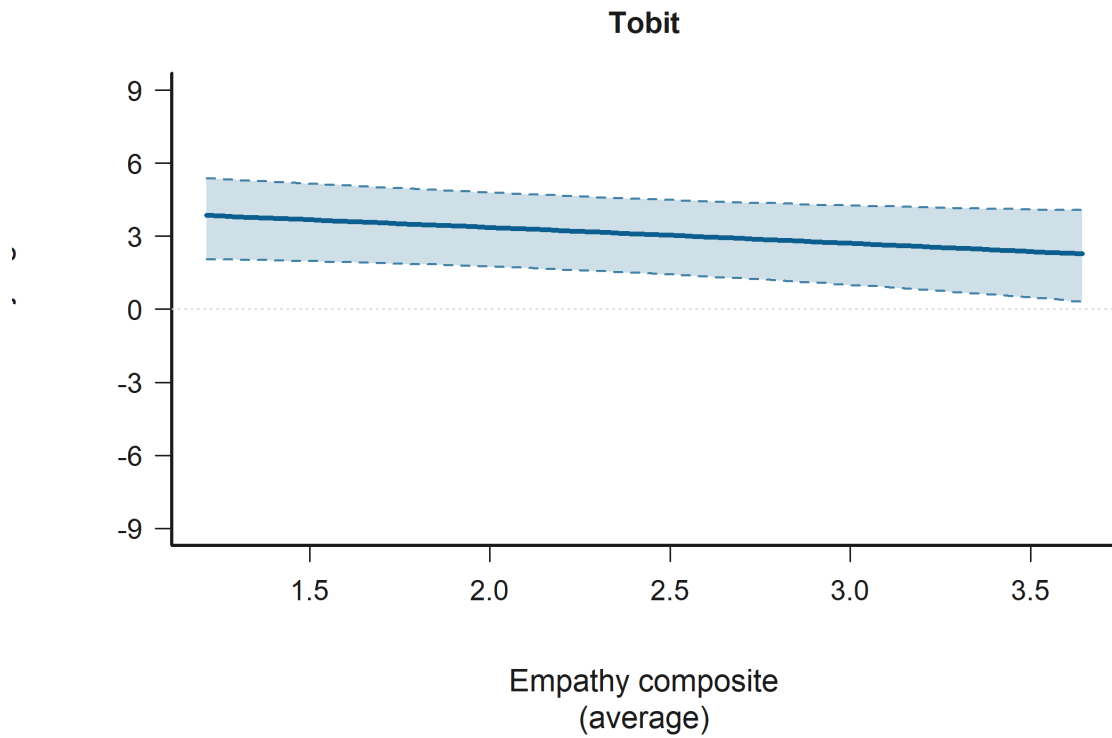


Figure 30: Effect Plot for Empathy composite (average) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Man (ref = woman) is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)

Figure 31 shows that predicted judgment is higher for Man than for Woman.

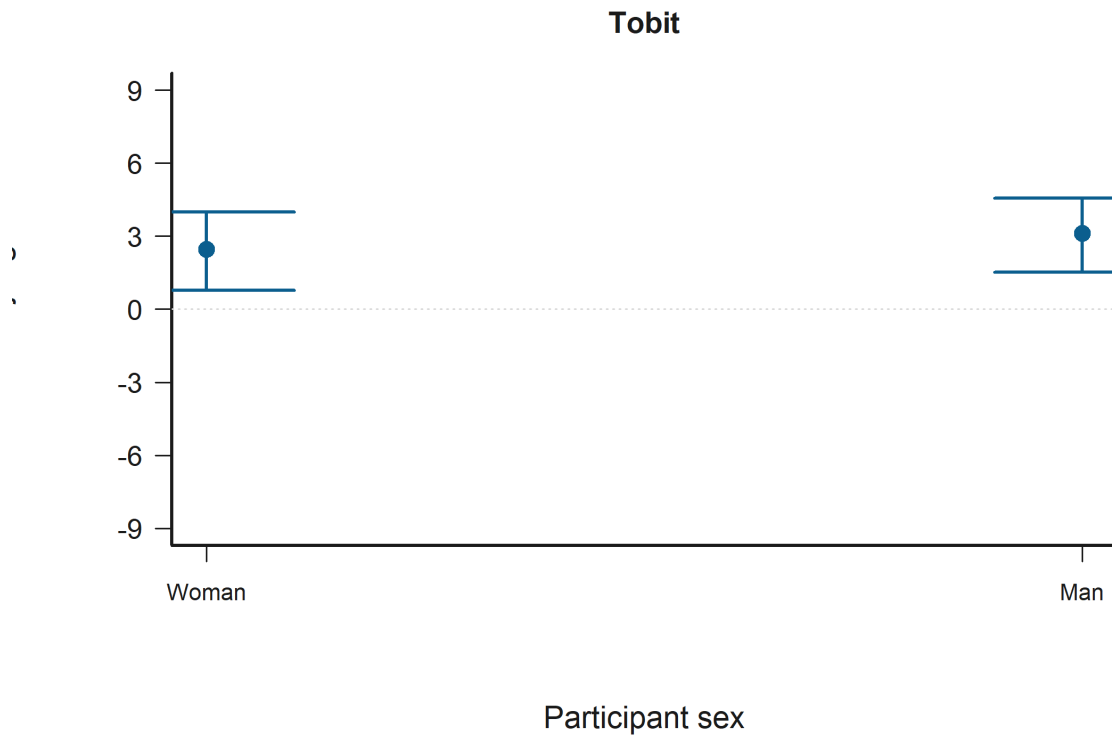


Figure 31: Grouped Prediction Plot for Man (ref = woman) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

judgments_victim Negotiator accepted harmful deal is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 32 shows that predicted judgment is lower for Accept harmful deal than for Reject harmful deal.



Figure 32: Grouped Prediction Plot for Negotiator accepted harmful deal in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Perspective taking is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 33 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

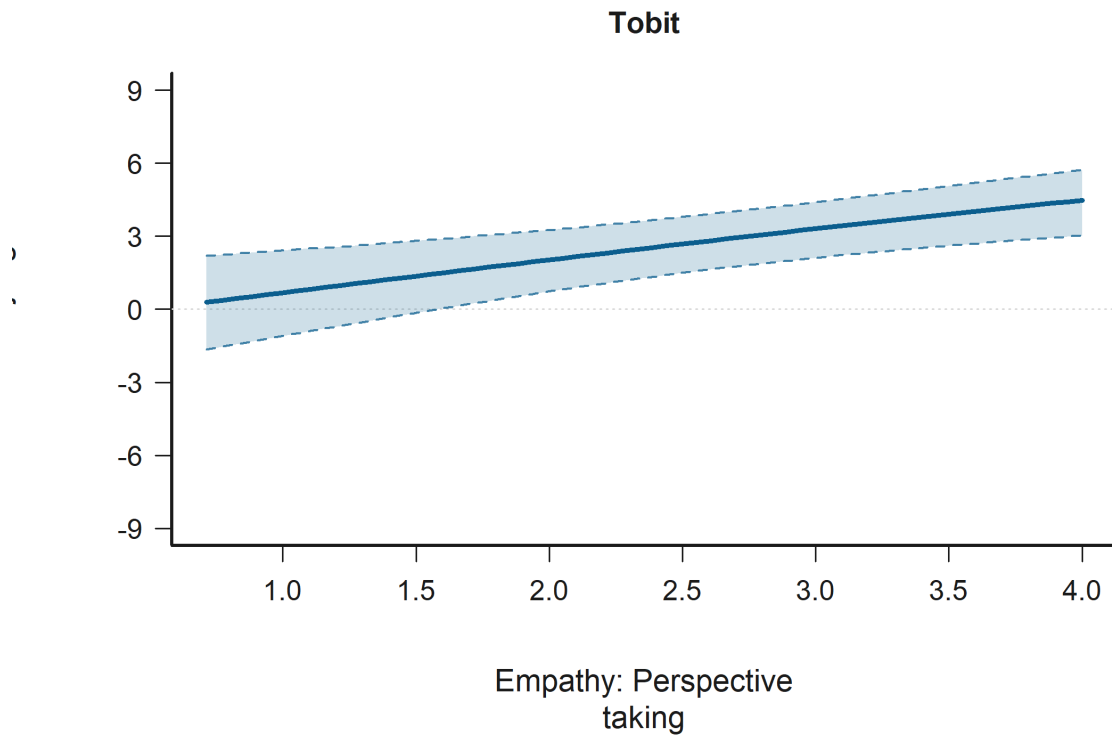


Figure 33: Effect Plot for Empathy: Perspective taking in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Empathic concern is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 34 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted judgment.

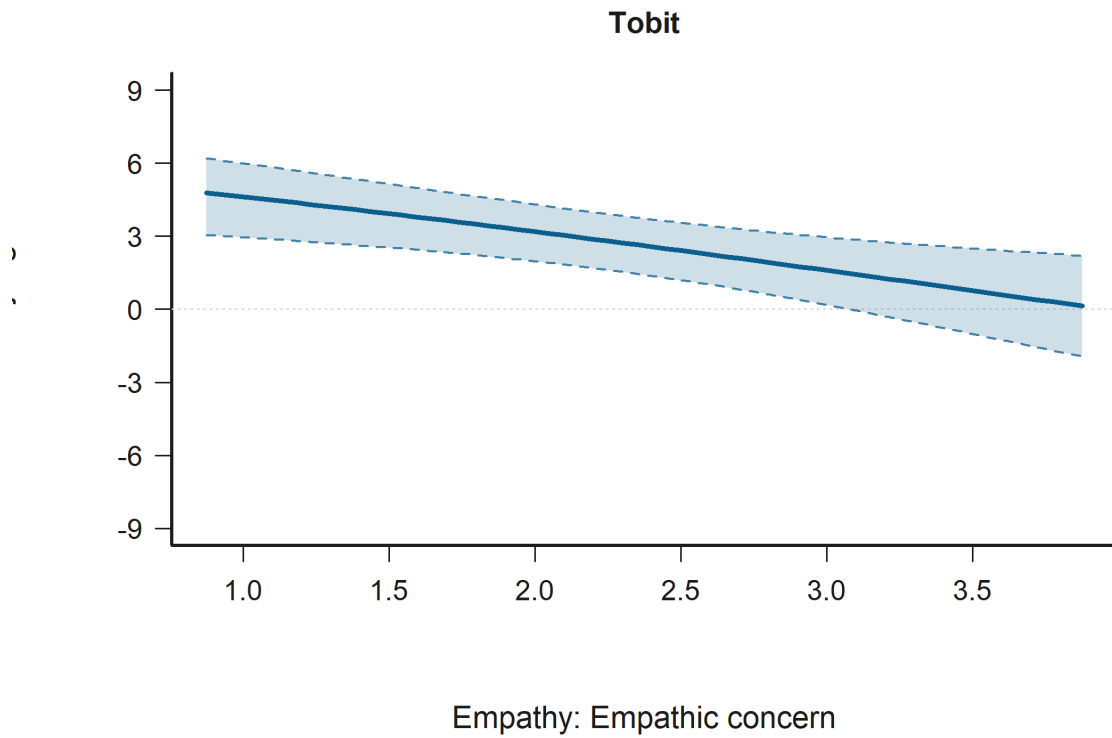


Figure 34: Effect Plot for Empathy: Empathic concern in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Socioeconomic status is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 35 shows that predicted judgment is higher for Socioeconomic status 5 than for Socioeconomic status 0.

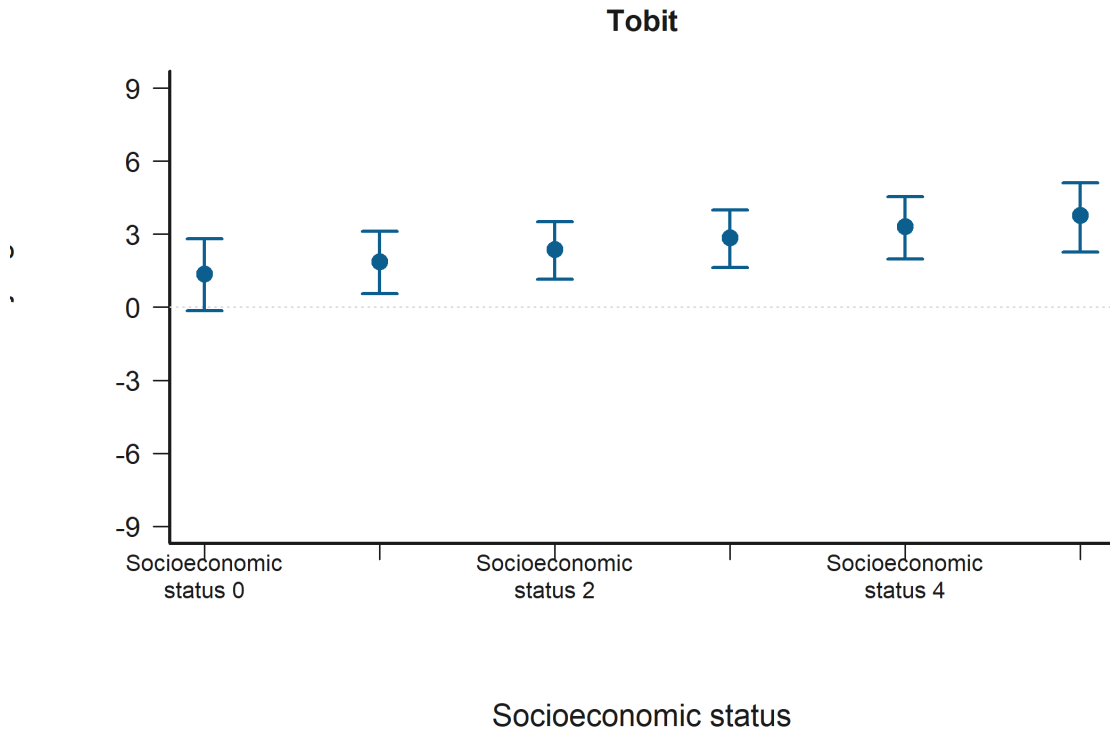


Figure 35: Grouped Prediction Plot for Socioeconomic status in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Judged negotiator outgroup (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 36 shows that predicted judgment is higher for Judged negotiator outgroup than for Judged negotiator ingroup.

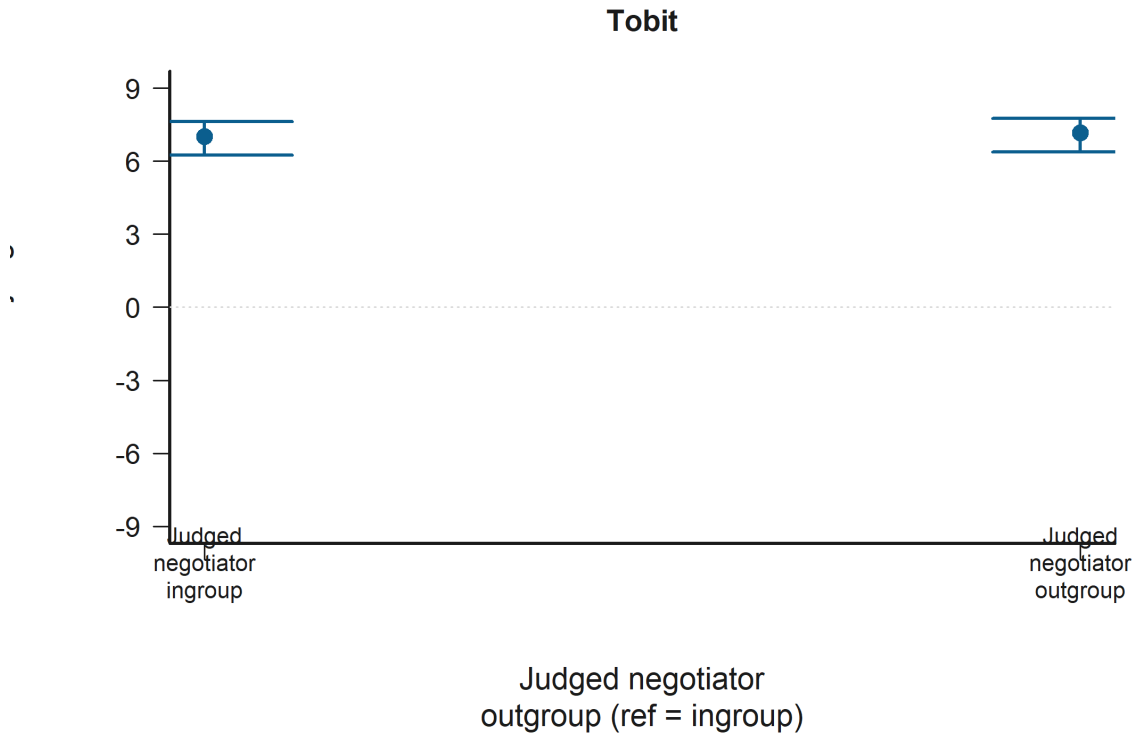


Figure 36: Grouped Prediction Plot for Judged negotiator outgroup (ref = ingroup) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Personal distress x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 37 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Judged negotiator outgroup.



Figure 37: Interaction Plot for Personal distress x judged negotiator outgroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathic concern x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 38 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Judged negotiator outgroup.

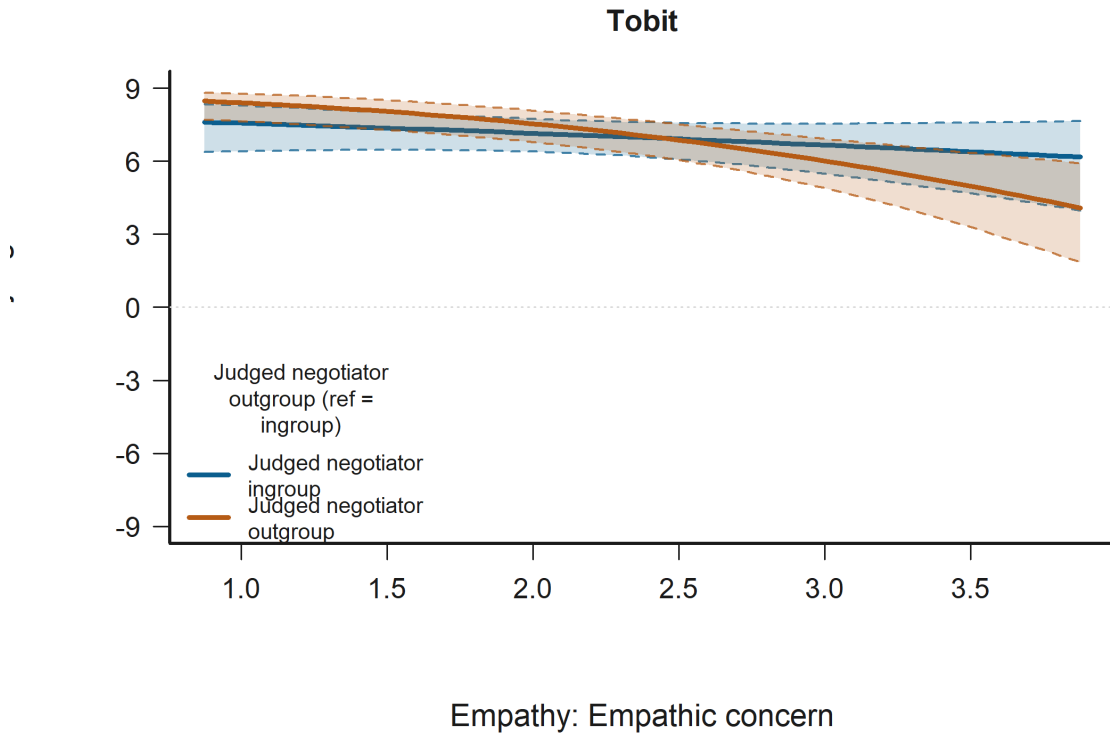


Figure 38: Interaction Plot for Empathic concern x judged negotiator outgroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Engineering participant (ref = humanities) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)
- H2: Negotiator-side relational structure Model A (Tobit)

Figure 39 shows that predicted judgment is higher for Engineering participant than for Humanities participant.

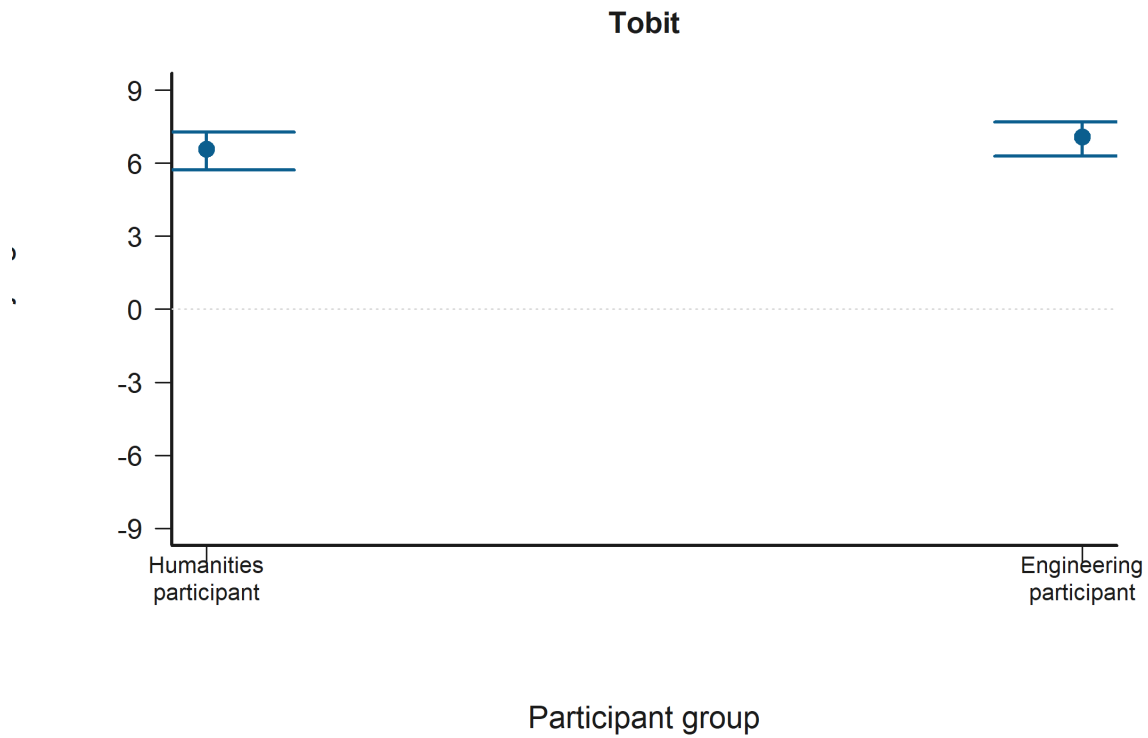


Figure 39: Grouped Prediction Plot for Engineering participant (ref = humanities) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy composite (average) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 40 shows that across the observed range, higher Empathy composite (average) corresponds to higher predicted judgment.

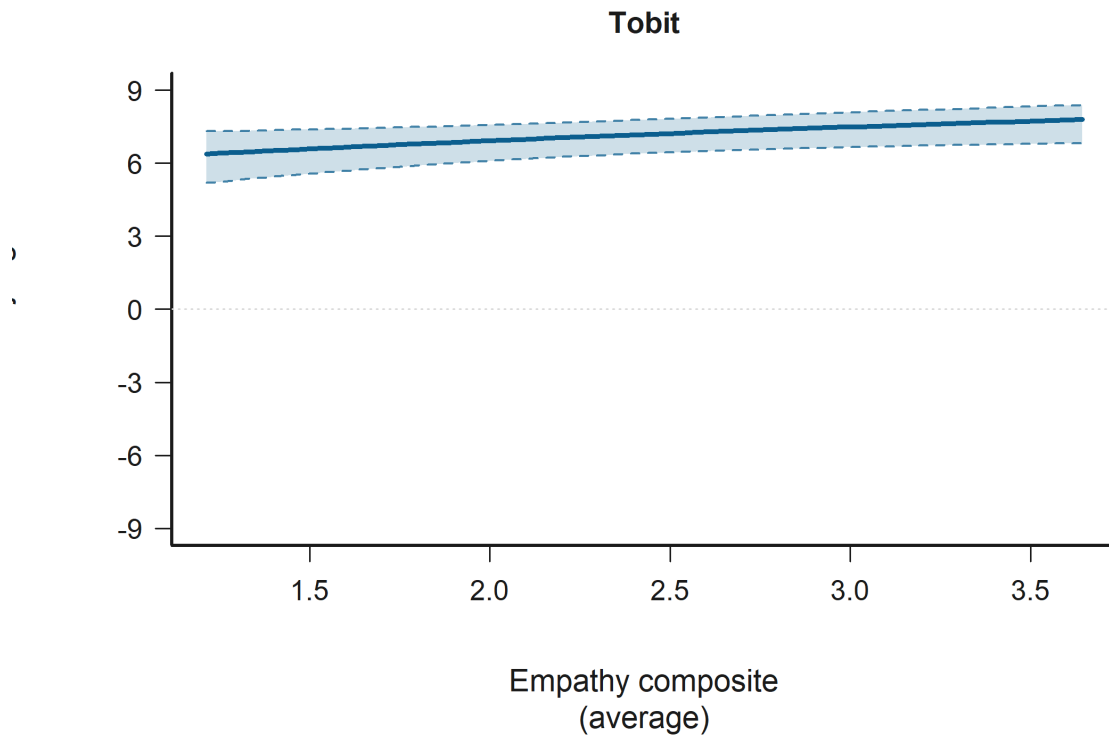


Figure 40: Effect Plot for Empathy composite (average) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy: Personal distress is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 41 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted judgment.

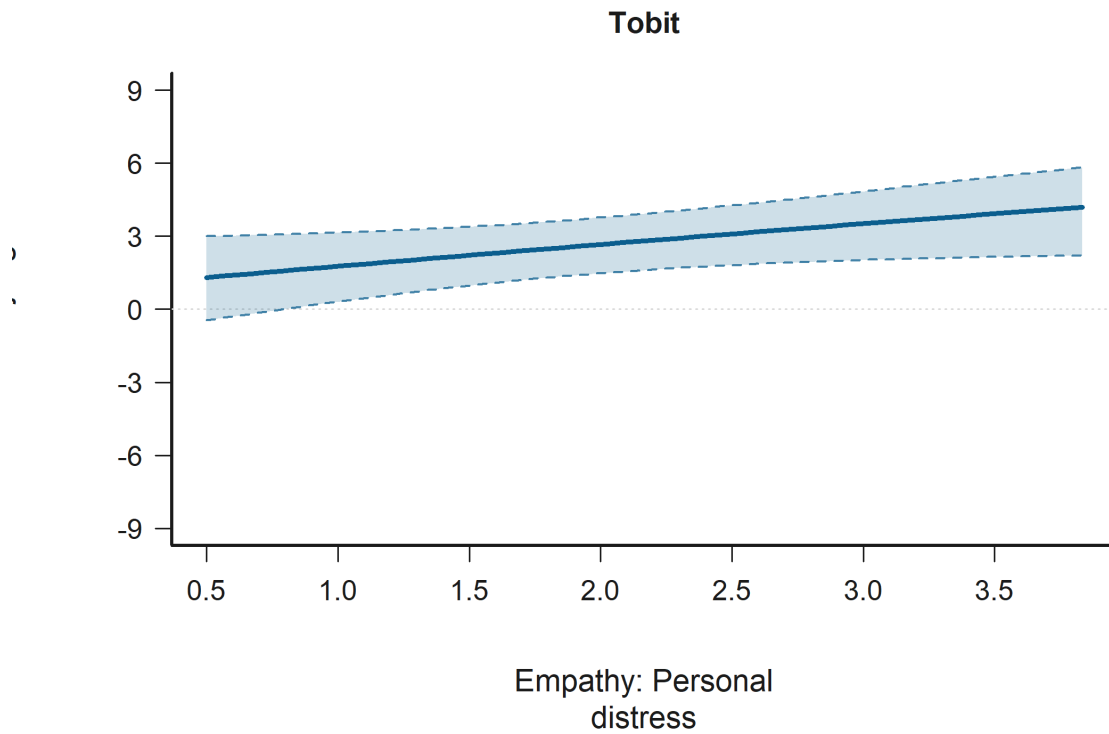


Figure 41: Effect Plot for Empathy: Personal distress in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 42 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is Judged negotiator ingroup.

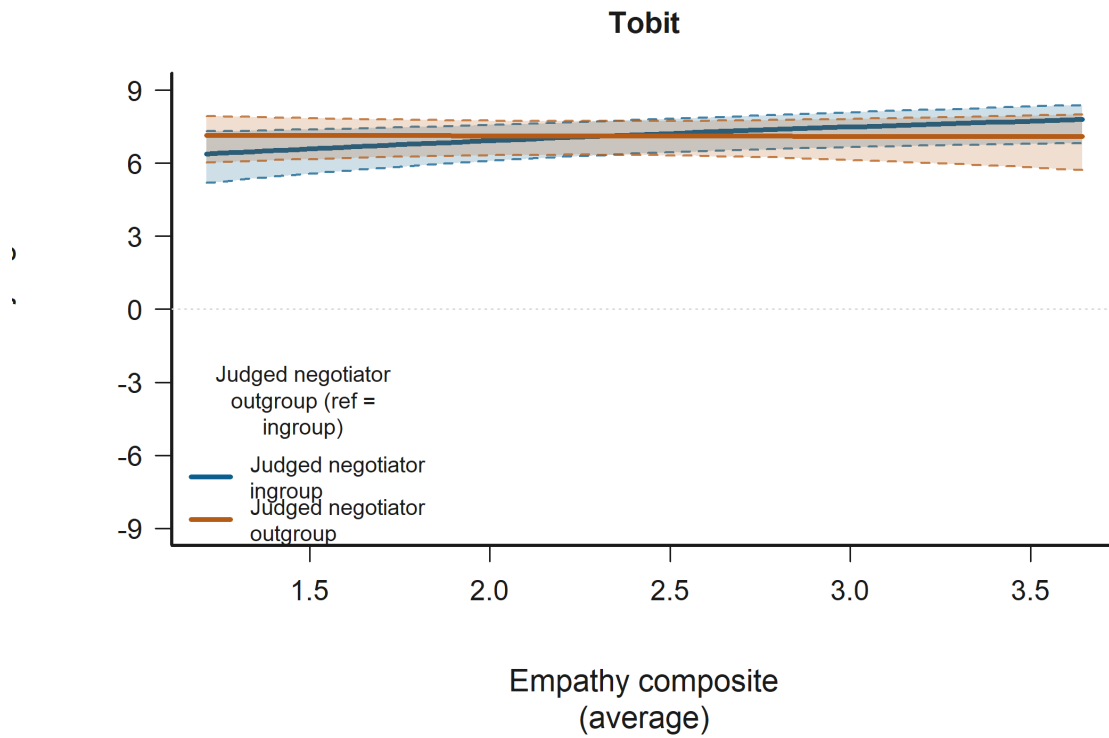


Figure 42: Interaction Plot for Empathy x judged negotiator outgroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Age is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H1: Empathy under relational controls Model A (Tobit)

Figure 43 shows that across the observed range, higher Age corresponds to higher predicted judgment.

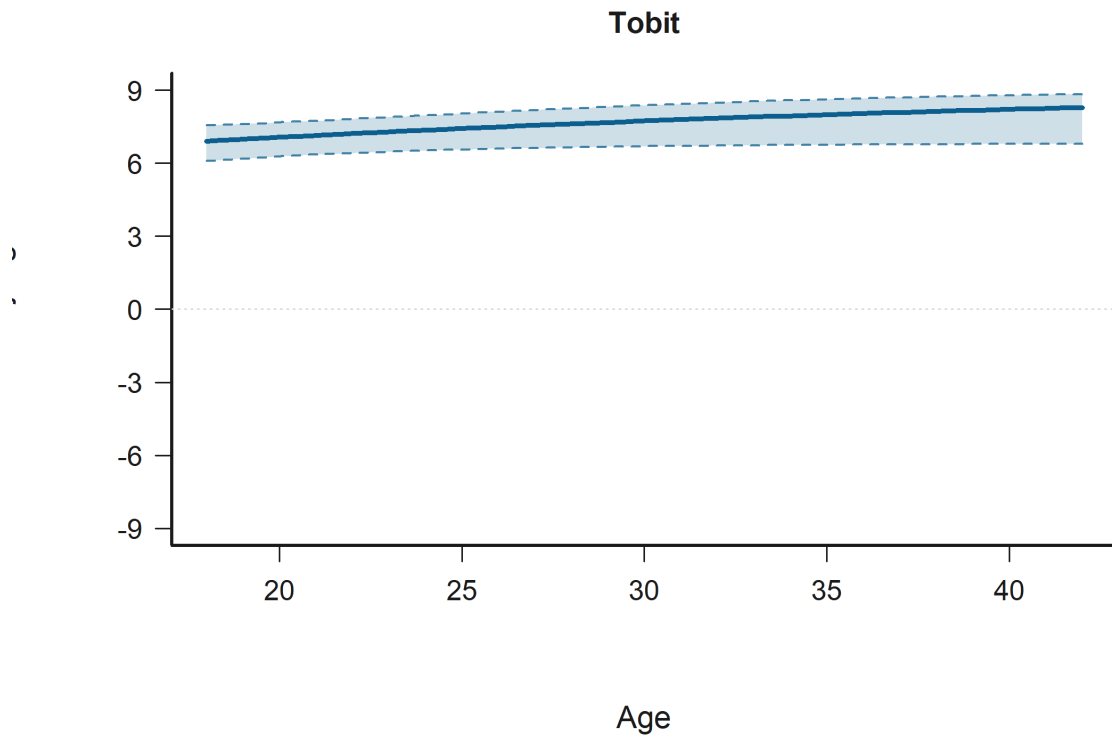


Figure 43: Effect Plot for Age in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Empathy x judged negotiator control label hidden is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 44 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is Judged negotiator ingroup.

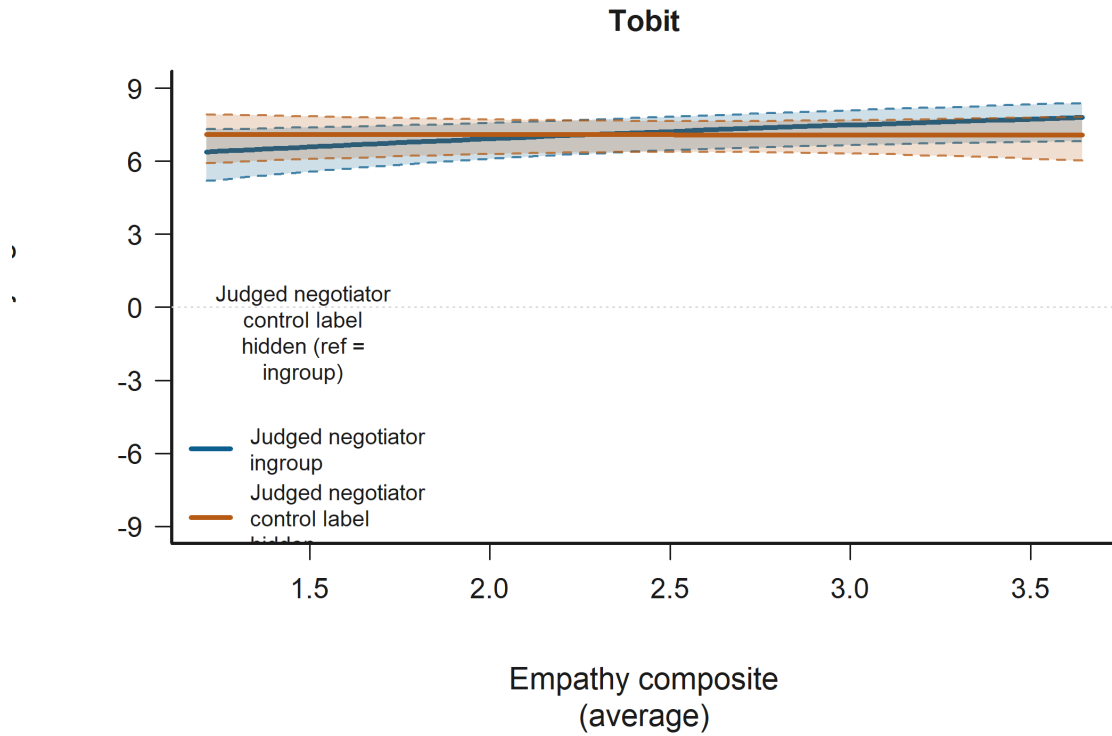


Figure 44: Interaction Plot for Empathy x judged negotiator control label hidden in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

Judged negotiator control label hidden (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 45 shows that predicted judgment is higher for Judged negotiator control label hidden than for Judged negotiator ingroup.



Figure 45: Grouped Prediction Plot for Judged negotiator control label hidden (ref = ingroup) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals.

8.4 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates support it in the current data.

- H1. Original hypothesis: Within the victim and bystander subsets, higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on judged-negotiator status, counterpart status, decision outcome, observer-side victim alignment when applicable, and participant controls. Victim subset: Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is positive but not statistically significant ($p = 0.417$). Model B supports the hypothesis through Empathy: Empathic concern with a negative association ($p = 0.004$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Engineering participant (ref = humanities) with a positive association ($p = 0.035$). Bystander subset: Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is negative but not statistically significant ($p = 0.800$). Model B supports the hypothesis through Empathy: Fantasy scale with a negative association ($p = 0.042$). Additional statistically significant signals include Negotiator accepted harmful deal with a

negative association ($p < 0.001$) and Engineering participant (ref = humanities) with a positive association ($p = 0.017$).

- H2. Original hypothesis: Moral-judgment severity should vary with the judgment-level ingroup/outgroup/control structure of the judged and counterpart negotiators. In the bystander subset, that negotiator-side structure should further depend on whether the player and the victim share faculty or not. Victim subset: Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the negotiator-side ingroup/outgroup/control structure dummies in the victim subset are statistically significant, and the closest signal is Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) with a negative association ($p = 0.319$). Model B does not support the hypothesis; none of the negotiator-side ingroup/outgroup/control structure dummies in the victim subset are statistically significant, and the closest signal is Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) with a negative association ($p = 0.298$). Additional statistically significant signals include Empathy: Perspective taking with a positive association ($p < 0.001$) and Empathy: Empathic concern with a negative association ($p = 0.003$). Bystander subset: Tobit conclusion: the available models support the hypothesis. Model A supports the hypothesis through Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim) with a positive association ($p = 0.035$). Model B supports the hypothesis through Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim) with a positive association ($p = 0.030$). Additional statistically significant signals include Socioeconomic status with a positive association ($p = 0.012$) and Empathy: Empathic concern with a negative association ($p = 0.028$).
- H3. Original hypothesis: The empathy effect may vary according to whether the judged negotiator is ingroup, outgroup, or control, while decision outcome and the additional relational controls remain explicitly modeled. Victim subset: Tobit conclusion: the available models support the hypothesis. Model A supports the hypothesis through Empathy x judged negotiator outgroup with a negative association ($p = 0.050$). Model B supports the hypothesis through Personal distress x judged negotiator outgroup with a positive association ($p = 0.026$) and Empathic concern x judged negotiator outgroup with a negative association ($p = 0.026$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Empathy: Perspective taking with a positive association ($p = 0.008$). Bystander subset: Tobit conclusion: the available models support the hypothesis. Model A supports the hypothesis through Empathy x judged negotiator control label hidden with a positive association ($p = 0.007$). Model B supports the hypothesis through Personal distress x judged negotiator outgroup with a positive association ($p = 0.016$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Judged negotiator control label hidden (ref = ingroup) with a negative association ($p = 0.013$).

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, including predictor estimates, standard errors, p-values, significance symbols, and estimator diagnostics for the victim and bystander subsets, are provided below.

9.1 H1: Empathy Effect

This section evaluates H1 separately in the victim and bystander subsets. Model A uses the overall empathy composite. Model B replaces that composite with the four IRI subscales: fantasy, empathic concern, perspective taking, and personal distress. Both models also retain judged-negotiator status, counterpart status, decision outcome, observer-side victim alignment when applicable, and participant controls for sex, age, and economic status.

9.1.1 Model A: Composite Empathy

Victim subset

Table 6: H1 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Victim subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	2.946	2.558	0.250	
Empathy composite (average)	0.491	0.605	0.417	
Judged negotiator outgroup (ref = ingroup)	0.014	0.456	0.975	
Judged negotiator control label hidden (ref = ingroup)	0.542	0.465	0.244	
Counterpart negotiator outgroup (ref = ingroup)	-0.548	0.432	0.204	
Counterpart negotiator control label hidden (ref = ingroup)	-0.596	0.460	0.195	
Observer-side victim outgroup (ref = ingroup victim)	NA	0.000	NA	
Negotiator accepted harmful deal	-14.951	0.960	<0.001***	***
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.210	0.573	0.035*	*
Man (ref = woman)	0.285	0.575	0.621	
Age	0.205	0.108	0.057+	+
Socioeconomic status	0.444	0.272	0.103	
Negotiator 2 (ref = negotiator 1)	0.352	0.342	0.303	
Scale variance (log)	1.994	0.055	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 2.946 ($p=0.250$). The term `iri_total` (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.491. The p-value of 0.417 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.014. The p-value of 0.975 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control` (representing whether the judged negotiator appeared in the control

or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.542. The p-value of 0.244 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_outgroup` (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.548. The p-value of 0.204 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_control` (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.596. The p-value of 0.195 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `observer_victim_outgroup` (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term `decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.951. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `role_observer` (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.210. The p-value of 0.035 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.285. The p-value of 0.621 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.205. The p-value of 0.057 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.444. The p-value of 0.103 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The `Log(scale)` is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter `log` is 1.994, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.940$, $p = 1.128e-27$). The primary Tobit model still uses participant-clustered standard errors by `id`.

Bystander subset

Table 7: H1 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Bystander subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	5.556	2.351	0.018*	*
Empathy composite (average)	-0.149	0.587	0.800	
Judged negotiator outgroup (ref = ingroup)	-0.205	0.386	0.595	
Judged negotiator control label hidden (ref = ingroup)	0.175	0.391	0.655	
Counterpart negotiator outgroup (ref = ingroup)	0.250	0.402	0.534	
Counterpart negotiator control label hidden (ref = ingroup)	0.852	0.391	0.029*	*
Observer-side victim outgroup (ref = ingroup victim)	-0.036	0.366	0.921	
Negotiator accepted harmful deal	-13.393	0.898	<0.001***	***
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.280	0.535	0.017*	*
Man (ref = woman)	-0.039	0.544	0.942	
Age	0.096	0.107	0.369	
Socioeconomic status	0.442	0.247	0.073+	+
Negotiator 2 (ref = negotiator 1)	0.342	0.322	0.289	
Scale variance (log)	1.896	0.050	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 5.556 ($p=0.018$). The term `iri_total` (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.149. The p-value of 0.800 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.205. The p-value of 0.595 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control` (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.175. The p-value of 0.655 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_outgroup` (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.250. The p-value of 0.534 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_control` (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.852. The p-value of 0.029 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `observer_victim_outgroup` (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.036. The p-value of 0.921 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.393. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `role_observer` (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient

of NA. The p-value of NA the p-value cannot be determined. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.280. The p-value of 0.017 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.039. The p-value of 0.942 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.096. The p-value of 0.369 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.442. The p-value of 0.073 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The `Log(scale)` is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter `log` is 1.896, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.941$, $p = 1.533e-27$). The primary Tobit model still uses participant-clustered standard errors by `id`.

9.1.2 Model B: Separated Empathy Constructs

Victim subset

Table 8: H1 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Victim subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	3.681	2.784	0.186	
Empathy: Fantasy scale	-0.183	0.551	0.740	
Empathy: Empathic concern	-2.302	0.796	0.004**	**
Empathy: Perspective taking	1.661	0.563	0.003**	**
Empathy: Personal distress	1.400	0.722	0.053+	+
Judged negotiator outgroup (ref = ingroup)	0.092	0.450	0.838	
Judged negotiator control label hidden (ref = ingroup)	0.459	0.460	0.319	
Counterpart negotiator outgroup (ref = ingroup)	-0.471	0.421	0.264	
Counterpart negotiator control label hidden (ref = ingroup)	-0.676	0.449	0.132	
Observer-side victim outgroup (ref = ingroup victim)	NA	0.000	NA	
Negotiator accepted harmful deal	-14.862	0.952	<0.001***	***
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.196	0.571	0.036*	*
Man (ref = woman)	0.163	0.610	0.790	
Age	0.170	0.109	0.120	
Socioeconomic status	0.300	0.264	0.255	
Negotiator 2 (ref = negotiator 1)	0.328	0.341	0.335	
Scale variance (log)	1.982	0.054	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 3.681 ($p=0.186$). The term `iri_fs` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.183. The p -value of 0.740 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_ec` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.302. The p -value of 0.004 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `iri_pt` (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.661. The p -value of 0.003 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `iri_pd` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.400. The p -value of 0.053 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.092. The p -value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control` (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.459. The p -value of 0.319 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_outgroup` (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.471. The p -value of 0.264 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_control` (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.676. The p -value of 0.132 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `observer_victim_outgroup` (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of NA. The p -value of NA the p -value cannot be determined. The term `decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.862. The p -value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `role_observer` (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p -value of NA the p -value cannot be determined. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.196. The p -value of 0.036 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.163. The p -value of 0.790 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.170. The p -value of 0.120 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic

contextual stratum of the participant’s background) has an estimated β coefficient of 0.300. The p-value of 0.255 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.982, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.943$, $p = 4.902\text{e-}27$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 9: H1 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Bystander subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	4.682	2.469	0.058+	+
Empathy: Fantasy scale	-1.067	0.524	0.042*	*
Empathy: Empathic concern	-0.841	0.800	0.293	
Empathy: Perspective taking	1.097	0.513	0.032*	*
Empathy: Personal distress	0.951	0.729	0.192	
Judged negotiator outgroup (ref = ingroup)	-0.125	0.384	0.745	
Judged negotiator control label hidden (ref = ingroup)	0.200	0.389	0.608	
Counterpart negotiator outgroup (ref = ingroup)	0.329	0.398	0.409	
Counterpart negotiator control label hidden (ref = ingroup)	0.870	0.392	0.027*	*
Observer-side victim outgroup (ref = ingroup victim)	-0.113	0.367	0.758	
Negotiator accepted harmful deal	-13.348	0.896	<0.001***	***
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.091	0.537	0.042*	*
Man (ref = woman)	0.069	0.563	0.902	
Age	0.092	0.108	0.393	
Socioeconomic status	0.408	0.242	0.092+	+
Negotiator 2 (ref = negotiator 1)	0.346	0.321	0.281	
Scale variance (log)	1.890	0.050	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 4.682 ($p=0.058$). The term *iri_fs* (representing the participant’s inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.067. The p-value of 0.042 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term *iri_ec* (representing the participant’s tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.841. The p-value of 0.293 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *iri_pt* (representing the participant’s tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.097. The p-value of 0.032 indicates statistically significant, meaning the predictor has a positive

effect on the latent moral judgment. The term `iri_pd` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.951. The p-value of 0.192 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.125. The p-value of 0.745 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control` (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.200. The p-value of 0.608 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_outgroup` (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.329. The p-value of 0.409 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_control` (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.870. The p-value of 0.027 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `observer_victim_outgroup` (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.113. The p-value of 0.758 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.348. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `role_observer` (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.091. The p-value of 0.042 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.069. The p-value of 0.902 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.092. The p-value of 0.393 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.408. The p-value of 0.092 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.890, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.945$, $p = 1.051e-26$). The primary Tobit model still uses participant-clustered standard errors by id,.

9.2 H2: Negotiator-Side Relational Structure

This section estimates H2 separately in the victim and bystander subsets. The dependent variable remains negotiator-specific judgment severity, so each participant contributes two judgment rows per scenario, one for each negotiator. In the victim subset, H2 uses the joint judged-plus-counterpart negotiator structure relative to the victim-player, with judged ingroup and counterpart ingroup as the reference structure. In the bystander subset, the same negotiator-side structure is defined relative to the observing player’s faculty, and the model additionally includes the player-victim outgroup term and its interaction with the negotiator-side structure dummies. Victim-subset equation: $y_{isj,Victim}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(V)} + \delta' \mathbf{Z}_i + \epsilon_{isj}$, where $\mathbf{S}_{isj}^{(V)}$ indexes the judged-plus-counterpart structure dummies and $\mathbf{Z}_i$ collects participant controls. Bystander-subset equation: $y_{isj,Obs}^* = \beta_0 + \beta_1 \text{Empathy}_i + \gamma' \mathbf{S}_{isj}^{(O)} + \eta V_{is} + \theta' (\mathbf{S}_{isj}^{(O)} \times V_{is}) + \delta' \mathbf{Z}_i + \epsilon_{isj}$, where V_{is} is the player-victim outgroup indicator.

9.2.1 Model A: Composite Empathy Control

Victim subset

Table 10: H2 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Victim subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	-2.604	3.639	0.474	
Empathy composite (average)	0.404	0.793	0.611	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.069	1.153	0.952	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-1.267	1.272	0.319	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.597	1.185	0.614	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-1.149	1.454	0.430	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.938	1.254	0.454	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.755	1.303	0.562	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-1.028	1.265	0.417	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.635	1.184	0.592	
Engineering participant (ref = humanities)	1.313	0.749	0.079	+
Man (ref = woman)	0.619	0.807	0.443	
Age	0.103	0.151	0.495	
Socioeconomic status	0.918	0.353	0.009**	**
Negotiator 2 (ref = negotiator 1)	0.286	0.497	0.565	
Scale variance (log)	2.445	0.060	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -2.604 (p=0.474). The term `iri_total` (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.404. The p-value of 0.611 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_in` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.069. The p-value of 0.952 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_in` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.267. The p-value of 0.319 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.597. The

p-value of 0.614 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.149. The p-value of 0.430 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.938. The p-value of 0.454 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.755. The p-value of 0.562 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.028. The p-value of 0.417 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.635. The p-value of 0.592 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.313. The p-value of 0.079 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.619. The p-value of 0.443 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.103. The p-value of 0.495 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.918. The p-value of 0.009 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The $\text{Log}(\text{scale})$ is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter $\log$ is 2.445, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.910$, $p = 8.459\text{e-}33$). The primary Tobit model still uses participant-clustered standard errors by id.

Bystander subset

Table 11: H2 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Bystander subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	-2.517	3.170	0.427	
Empathy composite (average)	-1.140	0.686	0.097	+
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-2.903	1.522	0.057	+
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	0.735	1.391	0.597	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.944	1.530	0.537	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.120	1.775	0.946	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	0.234	1.664	0.888	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.411	1.521	0.787	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.740	1.664	0.657	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	0.815	1.663	0.611	
Player-victim outgroup (ref = ingroup victim)	-2.378	1.639	0.147	
Engineering participant (ref = humanities)	1.146	0.628	0.068	+
Man (ref = woman)	0.994	0.678	0.143	
Age	0.271	0.125	0.029	*
Socioeconomic status	0.797	0.216	0.012	*
Negotiator 2 (ref = negotiator 1)	0.676	0.500	0.176	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	3.271	2.305	0.136	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	2.282	2.023	0.259	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	4.438	2.106	0.035	*
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	3.686	2.659	0.166	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup, counterpart negotiator outgroup (ref = ingroup victim)	-1.200	2.197	0.585	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	-0.434	2.204	0.844	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	0.728	2.134	0.733	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) x Player-victim outgroup (ref = ingroup victim)	1.657	2.147	0.440	
Scale variance (log)	2.309	0.061	< 0.001	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -2.517 ($p=0.427$). The term `iri_total` (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.140. The p-value of 0.097 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_in` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -2.903. The p-value of 0.057 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_in` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.735. The p-value of 0.597 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.944. The p-value of 0.537 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.120. The p-value of 0.946 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.234. The p-value of 0.888 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.411. The p-value of 0.787 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_cont` (representing the H2 negotiator-side structure contrast comparing

judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.740. The p-value of 0.657 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.845. The p-value of 0.611 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `player_victim_outgroup` (representing whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -2.378. The p-value of 0.147 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.146. The p-value of 0.068 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.994. The p-value of 0.143 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.274. The p-value of 0.029 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.797. The p-value of 0.012 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `h2_negstruct_j_out_c_in:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 3.271. The p-value of 0.156 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_in:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 2.282. The p-value of 0.259 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_out:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 4.438. The p-value of 0.035 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a discrete-by-discrete interaction, the positive coefficient indicates that the penalty or reward associated with one condition is lessened (more positive) when combined with the other condition. Note that any non-significant main effects involved in this term are superseded by this significant interaction. The term `h2_negstruct_j_out_c_out:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties)

has an estimated β coefficient of 3.686. The p-value of 0.166 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_out:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -1.200. The p-value of 0.585 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_cont:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -0.434. The p-value of 0.844 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_cont:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 0.728. The p-value of 0.733 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_cont:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 1.657. The p-value of 0.440 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The $\text{Log}(\text{scale})$ is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter $\log$ is 2.309, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.939$, $p = 8.609\text{e-}28$). The primary Tobit model still uses participant-clustered standard errors by `id`.

9.2.2 Model B: Separated Construct Controls

Victim subset

Table 12: H2 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Victim subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	-1.898	3.710	0.609	
Empathy: Fantasy scale	-0.511	0.720	0.478	
Empathy: Empathic concern	-2.969	0.998	0.003**	**
Empathy: Perspective taking	2.422	0.730	<0.001***	***
Empathy: Personal distress	1.655	0.810	0.049*	*
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	0.154	1.133	0.892	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-1.312	1.259	0.298	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.396	1.163	0.734	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-1.016	1.429	0.477	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.793	1.241	0.523	
Negotiator-side structure: judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.798	1.275	0.531	
Negotiator-side structure: judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.885	1.244	0.477	
Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)	-0.861	1.148	0.453	
Engineering participant (ref = humanities)	1.309	0.733	0.074 +	+
Man (ref = woman)	0.454	0.810	0.589	
Age	0.053	0.154	0.732	
Socioeconomic status	0.713	0.336	0.034*	*
Negotiator 2 (ref = negotiator 1)	0.269	0.497	0.588	
Scale variance (log)	2.437	0.060	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -1.898 ($p=0.609$). The term `iri_fs` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.511. The p-value of 0.478 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_ec` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.969. The p-value of 0.003 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `iri_pt` (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.422. The p-value of 0.001 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `iri_pd` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.655. The p-value of 0.049 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `h2_negstruct_j_out_c_in` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.154. The p-value of 0.892 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_in` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.312. The p-value of 0.298 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.396. The p-value of 0.734 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.016. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_out` (representing the

H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.793. The p-value of 0.523 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.798. The p-value of 0.531 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.885. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.861. The p-value of 0.453 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.309. The p-value of 0.074 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.454. The p-value of 0.589 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.053. The p-value of 0.732 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.713. The p-value of 0.034 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The $\text{Log}(\text{scale})$ is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter `log` is 2.437, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.923$, $p = 8.811\text{e-}31$). The primary Tobit model still uses participant-clustered standard errors by id.

Bystander subset

mated β coefficient of 0.259. The p-value of 0.882 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_out` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.479. The p-value of 0.775 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.260. The p-value of 0.865 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.492. The p-value of 0.769 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_cont` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.862. The p-value of 0.603 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `player_victim_outgroup` (representing whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -2.514. The p-value of 0.125 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.941. The p-value of 0.130 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 1.136. The p-value of 0.100 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.268. The p-value of 0.036 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.713. The p-value of 0.017 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `h2_negstruct_j_out_c_in:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 3.419. The p-value of 0.141 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_in:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 2.227. The p-value of 0.272 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_out:player_victim_outgroup` (representing

the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 4.576. The p-value of 0.030 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a discrete-by-discrete interaction, the positive coefficient indicates that the penalty or reward associated with one condition is lessened (more positive) when combined with the other condition. Note that any non-significant main effects involved in this term are superseded by this significant interaction. The term `h2_negstruct_j_out_c_out:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 3.513. The p-value of 0.185 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_out:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -1.307. The p-value of 0.553 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_in_c_cont:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -0.484. The p-value of 0.826 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_out_c_cont:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 0.612. The p-value of 0.776 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `h2_negstruct_j_cont_c_cont:player_victim_outgroup` (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 1.900. The p-value of 0.381 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.303, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.943$, $p = 5.932e-27$). The primary Tobit model still uses participant-clustered standard errors by id,.

9.3 H3: Empathy x Judged-Status Moderation

This section tests whether empathy slopes differ across judged-negotiator status categories after retaining decision outcome, judged-status x decision terms, counterpart status, and observer-side victim alignment.

9.3.1 Model A: Composite Empathy Interaction

Victim subset

Table 14: H3 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Victim subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	0.553	2.944	0.851	
Empathy composite (average)	1.602	0.753	0.033*	*
Judged negotiator outgroup (ref = ingroup)	3.870	1.947	0.047*	*
Judged negotiator control label hidden (ref = ingroup)	3.722	2.292	0.104	
Negotiator accepted harmful deal	-15.178	1.089	<0.001***	***
Counterpart negotiator outgroup (ref = ingroup)	-0.533	0.427	0.213	
Counterpart negotiator control label hidden (ref = ingroup)	-0.531	0.454	0.242	
Observer-side victim outgroup (ref = ingroup victim)	NA	0.000	NA	
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.214	0.570	0.033*	*
Man (ref = woman)	0.298	0.572	0.602	
Age	0.205	0.107	0.054+	+
Socioeconomic status	0.439	0.270	0.104	
Negotiator 2 (ref = negotiator 1)	0.367	0.341	0.281	
Accepted harmful deal x judged negotiator outgroup	-0.308	0.908	0.734	
Accepted harmful deal x judged negotiator control label hidden	0.856	0.925	0.355	
Empathy x judged negotiator outgroup	-1.664	0.849	0.050*	*
Empathy x judged negotiator control label hidden	-1.637	0.938	0.081+	+
Scale variance (log)	1.993	0.054	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 0.553 (p=0.851). The term `iri_total` (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 1.602. The p-value of 0.033 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 3.870. The p-value of 0.047 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `judged_control` (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 3.722. The p-value of 0.104 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -15.178. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term

counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.533. The p-value of 0.213 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.531. The p-value of 0.242 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.214. The p-value of 0.033 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.298. The p-value of 0.602 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.205. The p-value of 0.054 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.439. The p-value of 0.104 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.308. The p-value of 0.734 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.856. The p-value of 0.355 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:judged_outgroup (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -1.664. The p-value of 0.050 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the negative coefficient indicates that the slope of empathy on moral judgment is steeper/more negative (more severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term iri_total:judged_control (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -1.637. The p-value of 0.081 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. Because this is a continuous-by-discrete interaction, the negative coefficient indicates that the slope of empathy on moral judgment is steeper/more negative (more severe) for the specified condition compared to the

baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.993, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.940$, $p = 1.280e-27$). The primary Tobit model still uses participant-clustered standard errors by id.

Bystander subset

Table 15: H3 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Bystander subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	7.433	2.671	0.005**	**
Empathy composite (average)	-1.003	0.728	0.168	
Judged negotiator outgroup (ref = ingroup)	-0.555	1.713	0.746	
Judged negotiator control label hidden (ref = ingroup)	-4.651	1.878	0.013*	*
Negotiator accepted harmful deal	-13.017	0.960	<0.001***	***
Counterpart negotiator outgroup (ref = ingroup)	0.222	0.404	0.583	
Counterpart negotiator control label hidden (ref = ingroup)	0.773	0.390	0.048*	*
Observer-side victim outgroup (ref = ingroup victim)	-0.017	0.363	0.963	
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.271	0.535	0.017*	*
Man (ref = woman)	-0.058	0.544	0.915	
Age	0.093	0.108	0.387	
Socioeconomic status	0.433	0.246	0.079+	+
Negotiator 2 (ref = negotiator 1)	0.297	0.322	0.356	
Accepted harmful deal x judged negotiator outgroup	-0.571	0.811	0.481	
Accepted harmful deal x judged negotiator control label hidden	-0.455	0.793	0.566	
Empathy x judged negotiator outgroup	0.285	0.731	0.696	
Empathy x judged negotiator control label hidden	2.274	0.837	0.007**	**
Scale variance (log)	1.893	0.050	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 7.433 ($p=0.005$). The term `iri_total` (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.003. The p-value of 0.168 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.555. The p-value of 0.746 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control` (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -4.651. The p-value of 0.013 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `decision_accept` (representing whether the judged negotiator

accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.017. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `counterpart_outgroup` (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.222. The p-value of 0.583 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_control` (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.773. The p-value of 0.048 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `observer_victim_outgroup` (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.017. The p-value of 0.963 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `role_observer` (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.271. The p-value of 0.017 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.058. The p-value of 0.915 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.093. The p-value of 0.387 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.433. The p-value of 0.079 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup:decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.571. The p-value of 0.481 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control:decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.455. The p-value of 0.566 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_total:judged_outgroup` (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.285. The p-value of 0.696 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_total:judged_control` (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 2.274. The p-value of 0.007 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the slope of empathy on moral judgment is more positive (less severe) for the specified condition compared to the baseline

condition. The individual main effects of the components are subsumed by this contextual relationship. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.893, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.941$, $p = 2.122\text{e-}27$). The primary Tobit model still uses participant-clustered standard errors by id,.

9.3.2 Model B: Separated Constructs Interaction

Victim subset

Table 16: H3 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Victim subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	0.381	3.285	0.908	
Empathy: Fantasy scale	-0.276	0.776	0.722	
Empathy: Empathic concern	-1.215	1.018	0.233	
Empathy: Perspective taking	2.410	0.906	0.008**	**
Empathy: Personal distress	0.886	0.831	0.287	
Judged negotiator outgroup (ref = ingroup)	5.172	2.004	0.010**	**
Judged negotiator control label hidden (ref = ingroup)	4.713	2.729	0.084+	+
Negotiator accepted harmful deal	-15.017	1.066	<0.001***	***
Counterpart negotiator outgroup (ref = ingroup)	-0.454	0.416	0.275	
Counterpart negotiator control label hidden (ref = ingroup)	-0.619	0.439	0.158	
Observer-side victim outgroup (ref = ingroup victim)	NA	0.000	NA	
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.196	0.568	0.035*	*
Man (ref = woman)	0.128	0.609	0.833	
Age	0.182	0.111	0.101	
Socioeconomic status	0.293	0.262	0.262	
Negotiator 2 (ref = negotiator 1)	0.315	0.335	0.347	
Accepted harmful deal x judged negotiator outgroup	-0.492	0.875	0.574	
Accepted harmful deal x judged negotiator control label hidden	0.783	0.898	0.383	
Fantasy x judged negotiator outgroup	-0.578	0.784	0.461	
Empathic concern x judged negotiator outgroup	-2.530	1.140	0.026*	*
Perspective taking x judged negotiator outgroup	-0.910	0.934	0.330	
Personal distress x judged negotiator outgroup	2.168	0.976	0.026*	*
Fantasy x judged negotiator control label hidden	0.651	1.058	0.538	
Empathic concern x judged negotiator control label hidden	-0.793	1.291	0.539	
Perspective taking x judged negotiator control label hidden	-1.197	1.154	0.300	
Personal distress x judged negotiator control label hidden	-0.531	1.010	0.599	
Scale variance (log)	1.976	0.053	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 0.381 ($p=0.908$). The term `iri_fs` (representing the participant's inclination to transpose themselves

imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.276. The p-value of 0.722 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *iri_ec* (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.215. The p-value of 0.233 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *iri_pt* (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.410. The p-value of 0.008 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term *iri_pd* (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.886. The p-value of 0.287 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *judged_outgroup* (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 5.172. The p-value of 0.010 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term *judged_control* (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 4.713. The p-value of 0.084 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *decision_accept* (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -15.017. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term *counterpart_outgroup* (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.454. The p-value of 0.275 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *counterpart_control* (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.619. The p-value of 0.158 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *observer_victim_outgroup* (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term *role_observer* (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p-value of NA the p-value cannot be determined. The term *participant_engineering* (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.196. The p-value of 0.035 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term *sex_man* (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.128. The p-value of 0.833 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *age* (representing the participant's biological age in years) has an estimated β coefficient of 0.182. The p-value of 0.101 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *economic_status* (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.293. The p-value of 0.262 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term *judged_outgroup:decision_accept* (representing

whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.492. The p-value of 0.574 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control:decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.783. The p-value of 0.383 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_fs:judged_outgroup` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.578. The p-value of 0.461 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_ec:judged_outgroup` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -2.530. The p-value of 0.026 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the negative coefficient indicates that the slope of empathy on moral judgment is steeper/more negative (more severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term `iri_pt:judged_outgroup` (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.910. The p-value of 0.330 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pd:judged_outgroup` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 2.168. The p-value of 0.026 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the slope of empathy on moral judgment is more positive (less severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term `iri_fs:judged_control` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.651. The p-value of 0.538 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_ec:judged_control` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.793. The p-value of 0.539 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pt:judged_control` (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator appeared in the control

or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -1.197. The p-value of 0.300 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pd:judged_control` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.531. The p-value of 0.599 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The `Log(scale)` is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter `log` is 1.976, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.943$, $p = 6.009e-27$). The primary Tobit model still uses participant-clustered standard errors by `id`.

Bystander subset

Table 17: H3 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Bystander subset: 1,990 observations from 199 participants).

Predictor	Estimate	Std. Error	p-value	Signif.
Intercept	6.082	2.849	0.033*	*
Empathy: Fantasy scale	-1.235	0.621	0.047*	*
Empathy: Empathic concern	-0.708	0.901	0.432	
Empathy: Perspective taking	1.162	0.656	0.077+	+
Empathy: Personal distress	0.168	0.775	0.829	
Judged negotiator outgroup (ref = ingroup)	0.374	1.989	0.851	
Judged negotiator control label hidden (ref = ingroup)	-4.543	2.043	0.026*	*
Negotiator accepted harmful deal	-13.008	0.960	<0.001***	***
Counterpart negotiator outgroup (ref = ingroup)	0.350	0.397	0.379	
Counterpart negotiator control label hidden (ref = ingroup)	0.811	0.389	0.037*	*
Observer-side victim outgroup (ref = ingroup victim)	-0.087	0.366	0.812	
Observer role (ref = victim)	NA	0.000	NA	
Engineering participant (ref = humanities)	1.084	0.537	0.044*	*
Man (ref = woman)	0.030	0.562	0.958	
Age	0.092	0.108	0.392	
Socioeconomic status	0.394	0.240	0.101	
Negotiator 2 (ref = negotiator 1)	0.322	0.323	0.319	
Accepted harmful deal x judged negotiator outgroup	-0.650	0.792	0.412	
Accepted harmful deal x judged negotiator control label hidden	-0.381	0.795	0.631	
Fantasy x judged negotiator outgroup	-0.083	0.740	0.911	
Empathic concern x judged negotiator outgroup	-0.741	0.851	0.384	
Perspective taking x judged negotiator outgroup	-0.873	0.821	0.287	
Personal distress x judged negotiator outgroup	1.882	0.783	0.016*	*
Fantasy x judged negotiator control label hidden	0.503	0.738	0.495	
Empathic concern x judged negotiator control label hidden	0.390	0.808	0.629	
Perspective taking x judged negotiator control label hidden	0.759	0.809	0.348	
Personal distress x judged negotiator control label hidden	0.539	0.742	0.468	
Scale variance (log)	1.884	0.049	<0.001***	***

Tobit estimator Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 6.082 ($p=0.033$). The term `iri_fs` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.235. The p -value of 0.047 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `iri_ec` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.708. The p -value of 0.432 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pt` (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.162. The p -value of 0.077 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pd` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.168. The p -value of 0.829 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup` (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.374. The p -value of 0.851 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control` (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -4.543. The p -value of 0.026 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.008. The p -value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term `counterpart_outgroup` (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.350. The p -value of 0.379 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `counterpart_control` (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.811. The p -value of 0.037 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `observer_victim_outgroup` (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.087. The p -value of 0.812 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `role_observer` (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of NA. The p -value of NA the p -value cannot be determined. The term `participant_engineering` (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.084. The p -value of 0.044 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term `sex_man` (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.030. The p -value of 0.958 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `age` (representing the participant's biological age in years) has an estimated β coefficient of 0.092. The p -value of 0.392 indicates the

effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `economic_status` (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.394. The p-value of 0.101 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_outgroup:decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.650. The p-value of 0.412 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `judged_control:decision_accept` (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.381. The p-value of 0.631 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_fs:judged_outgroup` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.083. The p-value of 0.911 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_ec:judged_outgroup` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.741. The p-value of 0.384 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pt:judged_outgroup` (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.873. The p-value of 0.287 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pd:judged_outgroup` (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 1.882. The p-value of 0.016 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the slope of empathy on moral judgment is more positive (less severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term `iri_fs:judged_control` (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.503. The p-value of 0.495 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_ec:judged_control` (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.390. The p-value of 0.629 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pt:judged_control`

(representing the participant’s tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.759. The p-value of 0.348 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term `iri_pd:judged_control` (representing the participant’s tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.539. The p-value of 0.468 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The $\text{Log}(\text{scale})$ is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter $\log$ is 1.884, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.947$, $p = 3.249\text{e-}26$). The primary Tobit model still uses participant-clustered standard errors by `id`.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, both inference strategies assume independence between participants, which may still be violated by broader shared institutional or contextual shocks. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. The non-parametric robustness branch reduces dependence on that assumption and adds participant-level cluster bootstrap inference after converged full-sample fits, but bootstrap summaries still carry Monte Carlo error and can be sensitive to convergence problems in difficult specifications. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference across both estimators.

11 Conclusion

Based on the combined interval-censored Tobit estimations and the cluster-aware non-parametric robustness workflow, empathy and relational judgment structure have been documented together under Option 2 through negotiator-level relational predictors rather than descriptive case labels.